

Demographic Divergence and the Rise of Exporter Participation*

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Abstract

Over the past four decades, U.S. labor-force growth and business formation have fallen, the share of establishments belonging to exporting firms has risen, and U.S. labor-force growth has fallen behind that of its trading partners. We study these facts in a two-country model with heterogeneous firms, sunk export costs, endogenous entry and exit, and international borrowing. In a symmetric world, slower labor-force growth leaves firms' export and exit decisions unchanged on a balanced growth path and raises participation only by shifting the population of firms from entrants toward incumbents. When labor-force growth differs across countries, relative market size and international borrowing add a second channel that works through the real exchange rate. In an accounting of U.S. data over 1978–2022, labor-force growth accounts for about a third of the movement in exporter participation and about half of the movement in entry, but for little of the movement in exports; trade frictions account for most of the rest. More than half of demography's effect on participation comes from the demographic gap with trading partners, which a symmetric model cannot capture, and more than half of that works through borrowing.

JEL codes: F12, F32, F41, J11, L11.

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1 Introduction

Over the past four decades the U.S. business sector has changed in two ways that seem to point in opposite directions. It has become less dynamic: the establishment entry rate fell from 14.2 percent in 1978 to 9.9 percent in 2019. It has also become more international: the share of establishments that belong to exporting firms rose from 14.0 percent in 1992, the first year for which the Census Bureau reports firms' trading status, to 18.3 percent in 2019. A third change unfolded alongside them. The U.S. labor force grew by 2.6 percent a year in 1978 and by 0.4 percent in 2019,¹ and since the mid-1980s it has grown more slowly than the labor force of U.S. trading partners. This paper argues that the three changes are connected. A slower labor force means fewer new firms, a population of firms tilted toward the incumbents that export, and, because the slowdown has been deeper at home than abroad, an external adjustment that makes exporting more profitable.

Two channels link a slower labor force to exporting. The first is the composition of firms, and the data point to it. Between 1992 and 2019, shifts in the distribution of establishments across firm-size groups account for 97 percent of the rise in participation, and shifts across firm-age groups for 66 percent; participation within groups changed much less. Exporting requires a sunk investment, so exporters are concentrated among large, established incumbents. When the labor force grows more slowly, fewer new firms are needed to employ it, and because entrants are small and rarely export, the population of firms shifts toward incumbents that have already paid to enter foreign markets. The second channel is international and exists only because demographic growth differs across countries. When the home labor force grows more slowly than its partners', the foreign market grows relative to the home market and, in our model, home households lend abroad. The trade surplus that lending requires comes with a weaker real exchange rate, which makes exporting more profitable.

We study both channels in a two-country general equilibrium model with heterogeneous firms, sunk and per-period export costs, endogenous entry and exit, and a single internationally traded bond. Our first result characterizes the composition channel. On a balanced growth path of a symmetric world economy, with log utility and with entry and export costs paid in labor, the productivity thresholds at which firms exit, start exporting, and stop exporting do not depend on the growth rate of the labor force. Faster labor-force growth lowers the price of future consumption, but it raises the future cost of the labor that firms employ, and spend to enter and to export, in the same proportion, so the rate at which firms discount their values is unchanged. Demography therefore changes participation only

¹We quote 2019, the last year before the pandemic, which moved entry, trade, and the labor force sharply. In 2022 the entry rate was 10.8 percent, the exporter share 17.5 percent, and labor-force growth 1.1 percent. The accounting of Section 6 covers 1978–2022.

through the composition of firms, the weight of incumbents relative to entrants, and not through any firm’s decision to export. Differences in labor-force growth across countries break this invariance, because they move relative market size and the real exchange rate.

The two channels differ sharply in their quantitative footprint. Feeding only the observed U.S. and partner labor-force paths into the model, with trade costs held at their 1965 levels, reproduces the long-run direction of both the decline in entry and the rise in exporter participation, but not the swings around them. In an additive (Shapley) decomposition of the fitted paths over 1978–2022 (Section 6), with each contribution measured by the slope of the counterfactual path on the observed one, labor-force growth accounts for about half of the path of entry (0.52) and about a third of the path of participation (0.32). The effect on entry is entirely domestic. In a symmetric world economy, in which the labor force of U.S. partners follows the U.S. path, the same decomposition assigns labor-force growth as much of the path of entry (0.52) but only 0.13 of the path of participation. Entry responds to a country’s own labor force; exporter participation responds also to its labor force relative to its partners’. A closed or symmetric model can therefore account for the demographic origins of declining entry, but it captures less than half of the demographic effect on exporting. Controlled transitions between balanced growth paths show why (Section 5). While the home labor force grows faster than its partner’s, the foreign market shrinks relative to home and home borrows abroad, and both lower participation; once home grows more slowly, borrowing turns into lending and raises it. The borrowing channel does not rely on the love-of-variety link between labor-force growth and per-capita income growth: under log utility the equilibrium in units of labor is the same without it (Appendix C).

Our main quantitative exercise accounts for U.S. trade and firm dynamics over 1978–2022. We calibrate the model to a 1978 cross-section and recover, year by year, two bilateral iceberg wedges and a common export-cost wedge that reproduce U.S. exports, exporter participation, and the trade balance; the decomposition above runs over these fitted paths. Besides its shares of entry and participation, labor-force growth accounts for little of the path of exports relative to consumption (0.13). Most of the movement in participation, and nearly all of the movement in exports, comes from trade frictions: falling trade costs push participation up, and a rising export-cost wedge holds most of that push back. The model as a whole reproduces about half of the decline in entry and little of the decline in exit, so other forces act on business dynamism as well. Demography is thus a first-order but not a dominant force behind which firms export, the main force in the model behind the trend decline in entry, and a minor force behind how much is traded. Adding the ingredients of the model one at a time, 34 percent of demography’s effect on participation comes from domestic composition, 23 percent from relative market size, and 33 percent from international

borrowing and the real exchange rate; the split of the iceberg wedges, which the closure sets, accounts for the remaining 10 percent.

A further result helps interpret the historical record: demography accounts for the long-run decline in entry but not for its medium-run fluctuations. Splitting the entry rate into a Hodrick–Prescott trend and the deviations from that trend, demography accounts for 55 percent of the trend and 5 percent of the deviations. The deviations are tied to the external account. In the data, entry is above its trend when the U.S. trade deficit is above its own (the correlation of the two deviations is -0.74). In the model, the trade-cost wedges, which drive the trade balance, account for 52 percent of the deviations, whereas the symmetric model, which has no trade imbalances, accounts for 13 percent.

Two further results complete the picture. The welfare effect of the observed labor-force path, relative to growth held at its 1965 rate through 2022, is small: 0.18 percent of consumption at home and -0.30 percent abroad (Section 7). The U.S. labor force grew faster than its 1965 rate in the 1970s and more slowly afterwards, and gains in the early decades offset a permanent loss that arrives late: per-capita consumption at home ends 4.8 percent lower. Across the nine Census divisions, the regional data agree with the model in sign, more clearly for entry than for participation, but they have too little identifying variation to test its magnitudes (Section 8).

Related literature. Our paper is closest to a recent literature on the demographic origins of declining business dynamism. [Hopenhayn et al. \(2022\)](#) show that the slowdown of U.S. labor-force growth accounts for much of the decline in the firm entry rate and, by shifting the distribution of firms toward older and larger firms, for part of the rise in industry concentration; they organize the mechanism around an accounting identity that links the entry rate to labor-force growth, the exit rate, and the growth of average firm size. [Karahan et al. \(2024\)](#) reach a similar conclusion for the start-up rate using variation in labor-supply growth across U.S. states driven by past fertility, and show that in a standard model with free entry the decisions of incumbent firms do not depend on labor-supply growth, which is absorbed entirely at the entry margin. [Peters and Walsh \(2024\)](#) study declining population growth in a model of creative destruction and trace its consequences for entry, firm size, concentration, and productivity growth, and [Inokuma and Sánchez \(2025\)](#) link slower population growth to slower growth of aggregate productivity through firm dynamics. [Hopenhayn et al. \(2025\)](#) place population growth alongside sectoral technology and trade in a multisector entry–exit model to study sectoral reallocation. These papers study closed economies, or trade across sectors without an export decision, and our model reproduces their central result: in our accounting, demography is the main force behind the trend decline in entry, and entry re-

sponds only to the domestic labor force. We show that the same demographic force has an export counterpart, and that most of it is invisible to a closed economy. The invariance of incumbents' decisions that [Karahana et al. \(2024\)](#) establish extends to the sunk-cost export decision in a symmetric world, so a closed or symmetric economy captures only the composition part of the demographic effect on exporting, about a third of it. The rest comes from differences in demographic growth across countries, which reach exporters through relative market size and the external account.

Research on heterogeneous firms and exporter dynamics emphasizes selection, persistence, and sunk export costs ([Melitz, 2003](#); [Dixit, 1989](#); [Das et al., 2007](#); [Ruhl and Willis, 2017](#); [Alessandria and Choi, 2007, 2014](#); [Alessandria et al., 2021b](#)); [Ghironi and Melitz \(2005\)](#) is the closest two-country dynamic antecedent, and [Alessandria et al. \(2021a\)](#) and [Bernard et al. \(2018\)](#) survey the facts and frameworks. We add labor-force growth, characterize the resulting balanced growth path, and show which parts of exporter participation respond to domestic and to relative demography. Consistent with [Lincoln and McCallum \(2018\)](#), who find that falling export entry costs do not explain the rise of U.S. exporting, our accounting needs no fall in the cost of exporting to account for it.

A third literature places demographics and external imbalances in international models. [Cai and Stoyanov \(2016\)](#) make age-dependent skills a source of comparative advantage, and [Sposi \(2022\)](#) and [Nie et al. \(2024\)](#) embed demographic transitions in multi-country trade models to account for global imbalances through household age structure and life-cycle saving. [Kehoe et al. \(2018\)](#) study how U.S. borrowing reshaped the sectoral composition of employment. Our borrowing channel arises from differences in labor-force growth under a single non-contingent bond, without life-cycle saving, and it acts on the export decision rather than on sectoral employment. Our accounting follows the wedge tradition of [Chari et al. \(2007\)](#), applied to trade by [Alessandria et al. \(2013\)](#) and [Eaton et al. \(2016\)](#), and our treatment of the post-sample path of the wedges follows [Kehoe et al. \(2018\)](#).

Outline. Section 2 presents the facts. Section 3 develops the model and its balanced growth path. Section 4 describes the calibration. Section 5 compares the symmetric and asymmetric economies in controlled transitions. Section 6 reports the historical accounting. Section 7 computes the welfare effect of the slowdown, Section 8 reports regional evidence, and Section 9 concludes.

2 Facts

Three empirical patterns motivate the modeling choices of Section 3. All series are constructed from public BDS, BEA, BLS, and World Bank sources; construction details are in Appendix A.

Firm entry has fallen while the exporter share has risen. Figure 1(a) plots the firm-side trends; as in the introduction, we quote 2019, the last year before the pandemic (footnote 1 gives the 2022 values). The establishment entry rate fell from 14.2 percent in 1978 to 9.9 percent in 2019. The exporter share rose from 14.0 percent in 1992, the first year of the BDS trade-status tabulations, to 18.3 percent in 2019.² The entry decline is a well-documented business-dynamism fact; its co-movement with a *rising* exporter share is not.

U.S. labor-force growth has fallen behind its partners’. Figure 1(b) plots the demographic inputs of the model: U.S. civilian labor-force growth and export-weighted employment growth in the fifteen largest U.S. export markets. U.S. growth fell from 2.6 percent in 1978 to 0.4 percent in 2019, and the slowdown was not common to the United States and its partners. U.S. growth exceeded partner growth by 0.46 percentage points a year on average over 1965–1984; the difference was -0.23 points over 1985–2004 and -0.58 points over 2005–2022. A symmetric model cannot see this divergence; our two-country model is built around it.

The rise in the exporter share reflects the composition of firms. The aggregate exporter share is an average of the exporter shares of groups of establishments, weighted by each group’s share of establishments. Its change between two years splits exactly into a between-group part, due to shifts in the weights, and a within-group part, due to changes in participation inside each group. The BDS reports trading status by firm age and by firm size, but never by both at once, so Table 1 decomposes the same 1992–2019 change twice: once over four firm-age groups and once over three firm-size groups. Shifts across size groups account for 97 percent of the rise, and shifts across age groups for 66 percent. The share of establishments in firms at least eleven years old rose from 49 to 62 percent, and the share in firms with 500 or more employees from 13 to 19 percent, while participation within these

²The BDS assigns a firm’s trading status to all of its establishments, so the exporter share is the share of establishments that belong to goods-exporting firms. The share of firms that export is lower, and it also rose, from 3.3 to 4.1 percent. The model has single-establishment firms and, as in the business-dynamism literature, we treat establishments as firms. Before 1992 the exporter share is constructed from earlier Census benchmarks without using contemporaneous trade flows (dashed in the figure; Appendix A).

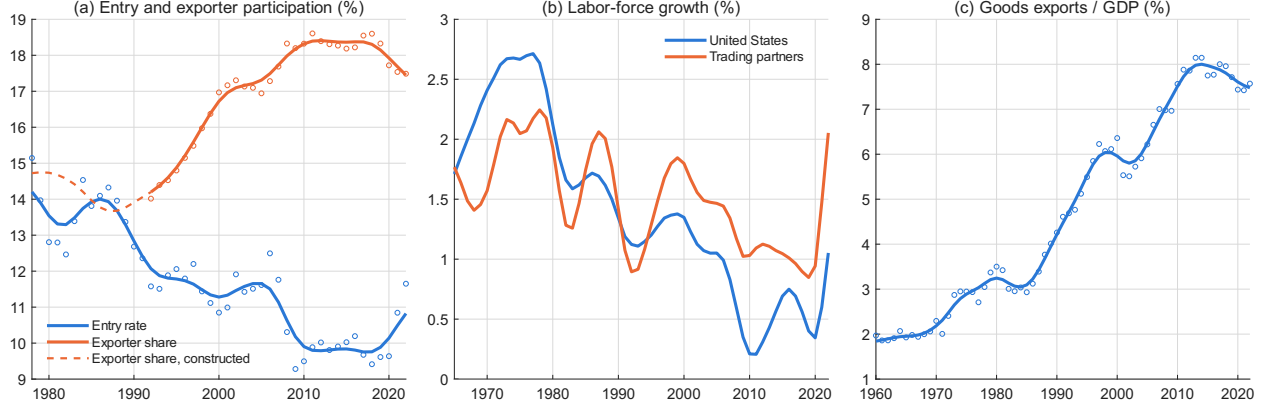


Figure 1: Entry, exporter participation, labor-force growth, and trade integration. Panel (a): establishment entry rate and share of establishments belonging to goods-exporting firms; the exporter share before 1992 is constructed. Panel (b): U.S. civilian labor-force growth and export-weighted employment growth in the fifteen largest U.S. export markets. Panel (c): U.S. real goods exports relative to real GDP. Markers are annual data and lines the smoothed series the model uses (Hodrick–Prescott, $\lambda = 6.25$).

Table 1: Decomposition of the 1992–2019 change in the exporter share (percentage points).

Decomposition over	Groups	Change	Between groups	Within groups	Between / change
Firm age	0, 1–5, 6–10, 11+	4.31	2.86	1.45	66%
Firm size	1–19, 20–499, 500+	4.31	4.18	0.13	97%

Notes: For groups g with establishment shares w_g and exporter shares p_g , the change in the aggregate exporter share is $\Delta P = \sum_g \bar{p}_g \Delta w_g + \sum_g \bar{w}_g \Delta p_g$, the between- and within-group parts, where bars denote averages of 1992 and 2019. Each row decomposes the same change over a different grouping of establishments; the BDS does not report trading status by age and size jointly, so the two rows cannot be combined. Firms born before 1977, whose exact age is not reported, are at least fifteen years old in 1992 and belong to the 11+ group. Over 1992–2022 the between-group shares are 68 percent (age) and 107 percent (size). Source: BDS Goods Traders.

groups moved little, from 25 to 28 percent and from 78 to 75 percent. The decompositions are descriptive, but both indicate that the rise in participation reflects which firms operate more than how often firms of a given type export.

From facts to model. Sunk export costs make exporter status a persistent state, so a shift of the firm population from entrants toward incumbents can raise aggregate participation without a broad rise in participation within groups. Country-specific labor-force growth and an international bond let relative demography act through relative market size, relative prices, and the external account. Finally, trade integration followed its own timing: real goods exports rose from 2.0 percent of real GDP in 1960 to a peak of 8.1 percent in 2014, accelerating in the late 1980s and 1990s while labor-force growth declined smoothly (Figure 1(c)). Time-varying trade and export-cost wedges absorb those movements, which

no slow-moving demographic series can match.

3 Model

There are two countries, home H and foreign F , and variables carry the country subscript $c \in \{H, F\}$. Time is discrete, a period is a year, and agents have perfect foresight. The labor force of country c evolves as

$$L_{c,t+1} = G_{L,c,t+1} L_{c,t}, \quad G_{L,c,t+1} \equiv 1 + n_{c,t}, \quad (1)$$

and $s_t \equiv L_{F,t}/L_{H,t}$ denotes relative size. We write G_L for a growth factor common to the two countries. The countries share all structural parameters; they differ in their labor-force paths, their initial conditions, and the wedges of Section 3.4. Each household member supplies one unit of labor inelastically; Section 3.9 relaxes this. The firm side follows [Alessandria and Choi \(2014\)](#). Firms have export status $e \in \{0, 1\}$. Each period a firm decides whether to operate or exit, and a firm that operates is hit by an exogenous exit shock with probability δ . Exporting requires a start-up cost and a continuation cost, with coefficients $\kappa_0 > \kappa_1 > 0$. The home final good is the numeraire, and the real exchange rate q_t is the price of the foreign final good in units of the home final good, so that a rise in q_t is a real depreciation of the home currency.

3.1 Households and the External Account

The representative household of country c maximizes

$$\sum_{t=0}^{\infty} \beta^t L_{c,t} \log c_{c,t}, \quad (2)$$

where $c_{c,t}$ is per-capita consumption of the country's final good and $\beta G_{L,c,t} < 1$, so that lifetime utility is finite. Households own the firms of their own country and trade a single one-period, non-contingent bond denominated in the home final good. Following [Schmitt-Grohé and Uribe \(2003\)](#), each household pays a premium that depends on its country's net foreign assets, which it takes as given: a household in country c buys the bond at the price $\bar{Q}_t e^{\psi \tilde{b}_{c,t+1}}$, where $\bar{Q}_t$ clears the bond market and $\tilde{b}_{c,t+1}$ is the country's net foreign assets divided by the value of its consumption in $t+1$. The premium is not a resource cost. The difference between what the two households pay per bond is rebated lump sum to the foreign household, with per-capita value R_t . With $b_{c,t}$ denoting per-capita bond holdings and $\Pi_{c,t}$

per-capita dividends, the budget constraints in home-good units are

$$c_{H,t} + G_{L,H,t+1} \bar{Q}_t e^{\psi \bar{b}_{H,t+1}} b_{H,t+1} = b_{H,t} + W_{H,t} + \Pi_{H,t}, \quad (3)$$

$$q_t c_{F,t} + G_{L,F,t+1} \bar{Q}_t e^{\psi \bar{b}_{F,t+1}} b_{F,t+1} = b_{F,t} + q_t (W_{F,t} + \Pi_{F,t}) + R_t, \quad (4)$$

where wages and dividends are measured in each country's own final good. Let $Q_{c,t} \equiv \beta c_{c,t}/c_{c,t+1}$ denote each country's discount factor in its own good. The first-order conditions are $\bar{Q}_t e^{\psi \bar{b}_{H,t+1}} = Q_{H,t}$ and $\bar{Q}_t e^{\psi \bar{b}_{F,t+1}} = Q_{F,t} q_t/q_{t+1}$, and eliminating $\bar{Q}_t$ gives the international Euler condition

$$Q_{H,t} e^{-\psi \bar{b}_{H,t+1}} = Q_{F,t} e^{-\psi \bar{b}_{F,t+1}} \frac{q_t}{q_{t+1}}. \quad (5)$$

Aggregating the home budget with goods-market clearing gives the external identity

$$Q_{H,t} B_{t+1} = B_t + TB_t, \quad TB_t \equiv EX_t - IM_t, \quad (6)$$

where $B_t = L_{H,t} b_{H,t}$ is home net foreign assets and bond-market clearing requires $L_{F,t} b_{F,t} = -B_t$. We use log utility because it is the case in which labor-force growth has no effect on the discount rate that firms apply (Section 3.6).

Balanced trade. We also solve the economy without the bond, the case the literature calls financial autarky. Then $B_t \equiv 0$, each country's absorption equals the value of its production, and trade is balanced, $TB_t = 0$, in every period; (5) does not apply. We refer to this economy as balanced trade. Comparing the two structures isolates how much of demography's effect on exporting travels through international borrowing (Sections 5 and 6).

3.2 Final-Good Producers

A representative producer in each country combines the varieties available in its market with a CES technology. Let $\tilde{\mu}_{c,t}(z, e)$ denote the measure of country- c firms that operate in period t with log-productivity z and export status e . The varieties available at home are those of all operating home firms and of the operating foreign firms that export, with mass $N_{H,t} = \sum_e \int \tilde{\mu}_{H,t}(z, e) dz + \int \tilde{\mu}_{F,t}(z, 1) dz$. Home final output is

$$D_{H,t} = N_{H,t}^{\nu - \frac{1}{\theta-1}} \left[\sum_e \int y_{h,t}(z, e)^{\frac{\theta-1}{\theta}} \tilde{\mu}_{H,t}(z, e) dz + \int y_{f,t}(z, 1)^{\frac{\theta-1}{\theta}} \tilde{\mu}_{F,t}(z, 1) dz \right]^{\frac{\theta}{\theta-1}}, \quad (7)$$

where y_h and y_f are the quantities of home and foreign varieties sold at home, $\theta > 1$ is the elasticity of substitution, and $\nu \in [0, 1/(\theta - 1)]$ the love-of-variety parameter of [Bénassy](#)

(1996); the foreign producer is symmetric. At the standard CES value $\nu = 1/(\theta - 1)$, which is our baseline, the leading factor equals one and a larger mass of varieties raises output per worker. Labor-force growth then raises the number of firms and, through variety, per-capita income. This is the scale effect of variety-based growth models (Romer, 1990; Sampson, 2016) in its semi-endogenous form, in which per-capita growth depends on the growth rate of the labor force rather than on its level (Jones, 1995, 2022). Under log utility this link has no effect on firms' decisions or on the external account; ν changes only the price of the final good in units of labor, and through it real consumption and the real exchange rate (Appendix C).

3.3 Intermediate Producers

Technology and static profits. A home firm with log-productivity z produces e^z units of its variety per unit of labor. Delivering one unit abroad requires shipping $(1 + \tau) \omega_{H,t}^\tau$ units, where τ is the calibrated structural iceberg cost and $\omega_{H,t}^\tau$ a time-varying wedge introduced in Section 3.4. With constant-elasticity demand, the firm charges a constant markup over marginal cost, $\frac{\theta}{\theta-1} W_{H,t} e^{-z}$ at home and $(1 + \tau) \omega_{H,t}^\tau$ times that price abroad, both in home goods. Static profit gross of fixed costs is

$$\pi_{H,t}(z, e) = e^{(\theta-1)z} (A_{HH,t} + e A_{HF,t}), \quad (8)$$

with destination-specific profit shifters, stated for the baseline ν ,

$$A_{HH,t} = \frac{1}{\theta} \left(\frac{\theta}{\theta-1} W_{H,t} \right)^{1-\theta} D_{H,t}, \quad A_{HF,t} = \frac{1}{\theta} \left(\frac{\theta}{\theta-1} (1 + \tau) \omega_{H,t}^\tau W_{H,t} \right)^{1-\theta} q_t^\theta D_{F,t}. \quad (9)$$

The export shifter depends on foreign absorption, the home wage, the iceberg cost of the home export leg, and the real exchange rate. A real depreciation raises $A_{HF,t}$ with elasticity θ ; this is the margin on which the external account acts. In each destination, revenue is $\theta/(\theta - 1)$ times the wage bill of the labor used in production. The foreign firm's problem is the mirror image in foreign goods, with shifters $A_{FF,t}$ and $A_{FH,t}$.

Dynamic problem. We describe the home firm and drop its country subscript. Within a period, an incumbent with state (z, e) first decides whether to operate; a firm that does not operate exits with value zero. An operating firm sells at home, and abroad if $e = 1$, pays the operating fixed cost $f_O W_t$, and chooses its export status for the next period, e' , paying $\omega_t^\kappa \kappa_e W_t$ if $e' = 1$. It then survives the exogenous exit shock with probability $1 - \delta$. Values are expressed relative to the per-capita output trend X_t (Section 3.6), and we write

$\hat{V} = V/X$ and $\hat{W} = W/X$. The value of an operating firm is

$$V_t^{\text{stay}}(z, e) = \hat{\pi}_t(z, e) - f_O \hat{W}_t + \max_{e' \in \{0,1\}} [-e' \omega_t^\kappa \kappa_e \hat{W}_t + \hat{\mathcal{V}}_t^{e'}(z)], \quad (10)$$

with continuation value

$$\hat{\mathcal{V}}_t^{e'}(z) = \mathcal{Q}_t(1 - \delta) \int \hat{V}_{t+1}(z', e') \phi(z' | z) dz', \quad \mathcal{Q}_t \equiv Q_{H,t} \frac{X_{t+1}}{X_t}, \quad (11)$$

where ϕ is the transition density of the Gaussian AR(1) process $z' = \rho_z z + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_z^2)$. Before deciding whether to operate, each firm draws an i.i.d. shock to the value of operating, logistic with mean zero and scale $\sigma_x \hat{W}_t$. The shock smooths the operating decision; it enters firm values but uses no resources. The value at the start of the period and the probability of operating are

$$\begin{aligned} \hat{V}_t(z, e) &= \hat{W}_t \sigma_x \log \left(1 + e^{V_t^{\text{stay}}(z, e)/(\hat{W}_t \sigma_x)} \right), \\ \iota_t(z, e) &= \frac{1}{1 + \exp \left(-V_t^{\text{stay}}(z, e)/(\hat{W}_t \sigma_x) \right)}. \end{aligned} \quad (12)$$

The model thus has two exit margins: the decision not to operate and the exogenous exit shock.

Entry. An unbounded pool of potential entrants can pay $f_E W_t$ at t to draw z from $\phi_E = \mathcal{N}(-\mu_E, \sigma_z^2/(1 - \rho_z^2))$, the ergodic distribution of the productivity process shifted down by μ_E . Entrants join the next period's population as non-exporters. Free entry requires

$$f_E \hat{W}_t = \mathcal{Q}_t \int \hat{V}_{t+1}(z', 0) \phi_E(z') dz' \quad (13)$$

whenever the mass of entrants M_t^E is positive. Entry and export costs are paid in labor. A labor-denominated entry cost makes the number of firms scale with the labor force rather than with productivity, the pattern that [Klenow and Li \(2025\)](#) document for U.S. firms.

3.4 Wedges

The accounting of Section 6 works with three time-varying wedges, each a multiplicative factor on a calibrated friction. The iceberg wedges $\omega_{H,t}^\tau$ and $\omega_{F,t}^\tau$ scale the iceberg cost of each export leg: shipping one unit from country c requires $d_{c,t} \equiv (1 + \tau) \omega_{c,t}^\tau$ units. The export-cost wedge ω_t^κ scales the start-up and continuation costs of both countries. A value of one reproduces the calibrated friction. The entry cost carries no wedge, so entry is an outcome of

the model rather than a target. The wedges are accounting devices: each absorbs whatever forces are needed to reproduce the series it is recovered from (Section 4). Trade in both directions restricts the product of the two iceberg wedges independently of relative prices and absorption, while their ratio is determined only jointly with the paths of relative prices and relative absorption, which depend on the closure of the external account (Section 3.7 and Appendix B). We therefore summarize trade costs by the geometric mean $\bar{\omega}_t^\tau \equiv (\omega_{H,t}^\tau \omega_{F,t}^\tau)^{1/2}$.

3.5 Distribution, Market Clearing, and Equilibrium

Let $\mu_{c,t}(z, e)$ be the measure of country- c incumbents at the start of period t , so that $\tilde{\mu}_{c,t} = \iota_{c,t} \mu_{c,t}$ is the measure of operating firms, and let $x_{c,t}(z, e)$ be the export policy. Write $\hat{\mu}_{c,t} = \mu_{c,t}/L_{c,t}$ for the per-capita measure and $m_{c,t} \equiv M_{c,t}^E/L_{c,t}$ for per-capita entry. Let $\mathcal{T}_{c,t}$ be the operator that maps a measure of incumbents into the measure of those that operate and move to (z', e') before the exit shock,

$$(\mathcal{T}_{c,t} \hat{\mu})(z', e') = \sum_e \int \iota_{c,t}(z, e) \mathbf{1}\{x_{c,t}(z, e) = e'\} \hat{\mu}(z, e) \phi(z' | z) dz.$$

The per-capita measure evolves as

$$\hat{\mu}_{c,t+1}(z', e') = \frac{1}{G_{L,c,t+1}} \left[(1 - \delta) (\mathcal{T}_{c,t} \hat{\mu}_{c,t})(z', e') + \mathbf{1}\{e' = 0\} m_{c,t} \phi_E(z') \right]. \quad (14)$$

Labor is used in production, operating costs, export costs, and entry. Writing $\ell_{c,t}^P$, $\ell_{c,t}^O$, and $\ell_{c,t}^X$ for the shares of the labor force used in the first three, computed over operating firms (Appendix B), labor-market clearing is

$$1 = \ell_{c,t}^P + \ell_{c,t}^O + \ell_{c,t}^X + f_E m_{c,t}. \quad (15)$$

The final good is used only for consumption, so $D_{c,t} = L_{c,t} c_{c,t}$. GDP equals the revenue of the country's firms, so per-capita GDP is $y_{c,t} = \frac{\theta}{\theta-1} W_{c,t} \ell_{c,t}^P$ in the country's own good, and home GDP equals $L_{H,t} c_{H,t} + TB_t$. We report the moments as their data counterparts are measured. The entry rate is M_t^E divided by the stock of firms at the start of the period, following the decision-period convention of Karahan et al. (2024). The exit rate counts all departures during the period, firms that choose not to operate plus operating firms hit by the exit shock, relative to the same stock. The exporter share is the fraction of operating firms that export.

Equilibrium. Given initial measures $\mu_{c,0}$, initial net foreign assets B_0 , and paths of the labor forces and the wedges, an equilibrium is a path of wages, the real exchange rate, and bond prices, together with consumption, bond holdings, firm values and policies, entry, and measures, such that: households maximize (2) subject to (3)–(4) and a no-Ponzi condition; firm values and policies solve (10)–(12); free entry (13) holds; the measures follow (14); the labor markets (15) and the markets for varieties and final goods clear; and the bond market clears. Appendix B lists the equilibrium conditions. The paths we compute end on a balanced growth path with common labor-force growth; Section 4 states the terminal conditions.

3.6 Balanced Growth with Labor-Force Growth

A balanced growth path is an equilibrium with constant wedges along which per-capita measures, per-capita entry, firm policies, expenditure shares, and the ratio of net foreign assets to consumption are constant, and per-capita variables grow at constant rates. Let $\Gamma_{H,t} \equiv A_{HF,t}/A_{HH,t}$ and $\Gamma_{F,t} \equiv A_{FH,t}/A_{FF,t}$ denote the relative export profitability of each country's firms: the profitability of selling abroad relative to selling at home.

Proposition 1 (Balanced growth). *Assume log utility and $\nu = 1/(\theta - 1)$. Then:*

- (i) *a balanced growth path with positive trade in both directions requires equal labor-force growth, $G_{L,H} = G_{L,F} = G_L$; the levels of the two labor forces may differ;*
- (ii) *along it the real exchange rate is constant, each country's wage satisfies $W_{c,t} \propto L_{c,t}^{1/(\theta-1)}$ in its own good, and per-capita output and consumption in both countries grow at*

$$G_p = G_L^{1/(\theta-1)}; \quad (16)$$

if $\psi > 0$, net foreign assets are zero;

- (iii) *the discount factor that firms apply to detrended values satisfies*

$$\mathcal{Q} = Q \frac{X_{t+1}}{X_t} = \frac{\beta}{G_p} \cdot G_p = \beta, \quad (17)$$

independently of G_L ;

- (iv) *with zero net foreign assets, G_L enters the detrended equilibrium conditions only through the factor $1/G_L$ in the two laws of motion (14);*
- (v) *the two free-entry conditions and an identity linking the two relative export profitabilities (Lemma 1, Appendix B) impose three restrictions, free of G_L , on the four statistics*

that summarize the aggregate state for firms, $(\hat{a}_{HH}, \Gamma_H, \hat{a}_{FF}, \Gamma_F)$, where $\hat{a}_{cc}$ is country c 's detrended domestic profit shifter divided by its wage. In a symmetric world these restrictions pin all four statistics, so both countries' export and exit cutoffs are independent of G_L . Away from symmetry they leave one dimension free, which is determined jointly with the rest of the equilibrium.

Part (iii) is the open-economy counterpart of an invariance known from closed economies: Karahan et al. (2024) show that with free entry incumbents' decisions do not depend on labor-supply growth, which is absorbed at the entry margin, and Hopenhayn et al. (2022) give conditions under which demography acts through the composition of firms. The open economy adds two things. The invariance covers the export decision, so the sunk-cost participation margin also moves only through composition; and with two countries it is exact only under symmetry. Part (i) is why every transition we compute converges to a common terminal growth rate: constant expenditure shares and a constant external position cannot coexist with a diverging relative size. Proofs are in Appendix B.

Corollary 1 (Composition). *On a symmetric balanced growth path, labor-force growth leaves firm policies unchanged and affects the firm population as follows:*

- (a) *the shape of the per-capita distribution depends on G_L only through the effective survival factor $(1 - \delta)/G_L$ in (14), while per-capita entry m scales the distribution without changing its shape;*
- (b) *the entry rate is strictly increasing in G_L ;*
- (c) *the share of the labor force used for entry, $f_E m$, is strictly increasing in G_L .*

Slower labor-force growth therefore acts on the firm population as higher survival would, with firms' decisions unchanged: incumbents gain weight relative to entrants, the entry rate falls, and less labor is devoted to creating firms. The model does not sign the effect on the exporter share in general, since it depends on how participation differs between incumbents and entrants. Entrants start as non-exporters and below the productivity of the average incumbent, and in our calibration the shift raises the exporter share (Section 5).

A natural conjecture is that slower labor-force growth lowers the interest rate, raises the option value of the sunk export cost, and draws more firms into exporting. Proposition 1(iii) shows that this conjecture fails on the balanced growth path under log utility. The per-capita growth that lowers the price of future consumption also raises the future wage in which fixed, entry, and export costs are paid, and the two effects cancel exactly. The cancellation rests on costs being paid in labor, and on the curvature of utility. With constant relative risk aversion σ , $\mathcal{Q} = \beta G_p^{1-\sigma}$, and the interest-rate channel is present with a strength governed by $\sigma - 1$:

at $\theta = 5$, lowering n from 2.64 percent to zero raises $\mathcal{Q}$ by 0.65 percent at $\sigma = 2$ and lowers it by 0.32 percent at $\sigma = 0.5$. Log utility is the case in which the channel is absent. Off the balanced growth path, $\mathcal{Q}_t$ moves with realized per-capita consumption growth relative to the trend, so the transitions we study are not exact channel decompositions.

3.7 Relative Demography, Relative Prices, and the External Account

When labor-force growth differs across countries, the invariance of Proposition 1(v) no longer holds. The following equilibrium relations describe how relative demography then reaches exporters. Let $\rho_t^c \equiv c_{H,t}/(q_t c_{F,t})$ and $\rho_t^y \equiv y_{H,t}/(q_t y_{F,t})$ denote home per-capita consumption and GDP relative to foreign, both in home goods, and $\rho_t^W \equiv W_{H,t}/(q_t W_{F,t})$ the relative wage. Let $\mathcal{Z}_{c,t} \equiv \sum_e \int e^{(\theta-1)z} \tilde{\mu}_{c,t}(z, e) dz$ be the productivity-weighted mass of country c 's operating firms and $\mathcal{Z}_{c,t}^x$ the same mass over its exporters.

Proposition 2 (Equilibrium relations with asymmetric demography). *Let $\nu = 1/(\theta - 1)$. Along any equilibrium path, on or off a balanced growth path:*

(i) *relative export profitability satisfies*

$$\Gamma_{H,t} = d_{H,t}^{1-\theta} q_t^{\theta-1} \frac{s_t}{\rho_t^c}, \quad \Gamma_{F,t} = d_{F,t}^{1-\theta} q_t^{1-\theta} \frac{\rho_t^c}{s_t}; \quad (18)$$

(ii) *the real exchange rate satisfies*

$$q_t^{1-\theta} = \frac{(\rho_t^W)^{\theta-1} \mathcal{Z}_{F,t} + d_{H,t}^{1-\theta} \mathcal{Z}_{H,t}^x}{\mathcal{Z}_{H,t} + d_{F,t}^{1-\theta} (\rho_t^W)^{\theta-1} \mathcal{Z}_{F,t}^x}; \quad (19)$$

(iii) *with the bond, relative consumption evolves as*

$$\rho_{t+1}^c = \rho_t^c \exp[-\psi(\tilde{b}_{H,t+1} - \tilde{b}_{F,t+1})]; \quad (20)$$

under balanced trade, $\rho_t^c = \rho_t^y$;

(iv) *the trade balance satisfies*

$$\frac{TB_t}{L_{H,t} c_{H,t}} = \frac{s_t (\rho_t^y - \rho_t^c)}{\rho_t^c (\rho_t^y + s_t)}, \quad (21)$$

so home runs a surplus if and only if $\rho_t^y > \rho_t^c$;

(v) *relative income satisfies*

$$\rho_t^y = \rho_t^W \frac{\ell_{H,t}^P}{\ell_{F,t}^P}, \quad \ell_{c,t}^P = 1 - \ell_{c,t}^O - \ell_{c,t}^X - f_E m_{c,t}. \quad (22)$$

Parts (i)–(ii) follow from the profit shifters and the two price indices, part (iii) from (5), part (iv) from the world resource constraint, and part (v) from markup pricing and (15) (Appendix B). These are equilibrium conditions, not comparative statics: relative size is the only exogenous object in them, and the real exchange rate, relative consumption, relative income, and the firm populations are determined jointly.

Relative size and relative prices. Part (i) decomposes the relative export profitability of home firms into trade costs, relative size, the real exchange rate, and relative per-capita absorption. Relative size enters with unit elasticity but does not move alone. By part (ii), the real exchange rate is the relative price of the two consumption bundles, and it depends on the relative wage and on the productivity-weighted firm populations of the two countries. As long as the two import shares sum to less than one, a proportional expansion of the foreign firm population at a given relative wage lowers q_t : a larger set of foreign varieties makes the foreign bundle cheaper, which lowers the profitability of home exports. A foreign labor force that grows faster than the home one therefore raises $\Gamma_{H,t}$ through s_t and lowers it through q_t . The net effect depends on the response of the relative wage and of the firm populations, and Section 5 measures it.

Borrowing. Parts (iii)–(v) describe the external account. With the bond, relative per-capita consumption changes only through the premium, and by part (iv) home runs a trade surplus exactly when its relative per-capita income exceeds its relative per-capita consumption. When $\psi = 0$ and initial net foreign assets are zero, ρ_t^c is constant and lies within the range of ρ_t^y along the path, so a country lends while its relative income is high and borrows while it is low. Under balanced trade relative consumption equals relative income in every period. Part (v) identifies what moves relative income: the relative wage, which moves with the firm populations through part (ii), and the share of labor used in production, which falls when more labor is used to create firms. By Corollary 1(c), the share of labor devoted to entry rises with labor-force growth on a balanced growth path. Along a transition in which one country's labor force grows faster than the other's for a time, the same force lowers that country's relative income and leads it to borrow. By part (i), the higher relative absorption of a borrowing country lowers the relative export profitability of its firms, and a lending country's lower relative absorption raises it. Under balanced trade this route is closed, and

relative prices move only as needed to balance trade.

3.8 The Symmetric Economy

When the two countries share the same labor-force path, the same wedges, and the same initial conditions with $B_0 = 0$, the equilibrium is symmetric: the bond is not traded, and $TB_t = 0$, $B_t = 0$, and $q_t = 1$ in every period. We call this special case the symmetric economy. It is the setting of Proposition 1(v) and Corollary 1, and it is the natural open-economy counterpart of the closed economies studied in the demography and business-dynamism literature. We use it throughout as the benchmark against which the asymmetric economy is compared: in it, demography acts only through the composition of firms.

3.9 Extensions of the Balanced-Growth Results

Endogenous labor supply. With King–Plosser–Rebelo preferences (King et al., 1988), the intratemporal condition $\chi h_t^\varphi c_t = W_t$ keeps hours h_t constant on the balanced growth path because consumption and the wage grow at the same rate, and (17) is unchanged (Appendix G).

The closed economy. Setting both export costs to zero, $\kappa_0 = \kappa_1 = 0$, removes the export decision. Every incumbent then exports, and only entrants, which start as nonexporters, do not. On the symmetric balanced growth path an incumbent’s profits are $e^{(\theta-1)z} A_{HH}(1 + \Gamma)$, with $\Gamma = d^{1-\theta}$ the relative export profitability of Proposition 2, and a firm in its first period earns $e^{(\theta-1)z} A_{HH}$. Up to that lower first-period profitability, the economy is therefore a closed economy with free entry, endogenous exit, and a common demand shifter, the class of economies studied by Hopenhayn et al. (2022) and Karahan et al. (2024). Demography acts only through entry, exit, and firm size, and Proposition 1 and Corollary 1 apply unchanged. Section 5.1 and Figure 3 use this limit, at the parameters of the open economy, to isolate what the export decision adds.

4 Calibration and Accounting

The model is disciplined in two steps. The structural parameters are set to match a 1978 cross-section of U.S. establishments, the first year of BDS coverage, on a balanced growth path with every wedge equal to one. Given these parameters and the two labor-force paths, the three wedges are then recovered year by year over 1965–2022 so that the model reproduces home exports, home exporter participation, and the trade balance. Section 6 reports the

accounting over 1978–2022; Appendix D shows the full paths, from 1965 to the end of the transition.

4.1 Structural Parameters

Table 2 reports the parameters. Five are assigned. The elasticity $\theta = 5$ implies a 25 percent markup (Broda and Weinstein, 2006). The initial labor-force growth rate, $n_0 = 2.64$ percent, is the 1978 trend growth rate of the U.S. civilian labor force. The discount factor $\beta = 0.966$ sets the bond price on the calibrated balanced growth path to 0.96; by Proposition 1(iii), firms discount detrended values at β . The scale of the operating shock is fixed at $\sigma_x = 0.001$ units of labor, 3.7 percent of the operating fixed cost; it smooths the operating decision and is not estimated. The structural iceberg cost $\tau = 0.78$ sets the share of output that an exporter sells abroad to 9.0 percent: on the symmetric balanced growth path every exporter sells the fraction $(1 + \tau)^{1-\theta} / [1 + (1 + \tau)^{1-\theta}]$ of its output abroad. The target interpolates linearly to 1978 the export intensity of U.S. manufacturing exporters reported by Bernard and Jensen (1995), 8.5 percent in 1976 and 11.4 percent in 1987.

The remaining seven parameters are chosen jointly to minimize a weighted distance between model and data moments: the exporter share, the stopper ratio (the fraction of last year’s surviving exporters that stop exporting; Bernard and Jensen, 1999, as used by Alessandria and Choi, 2007), the entry and exit rates, the employment shares of entering and exiting establishments, and the distributions of establishments and of employment across seven establishment-size classes (Figure 2). Model firms correspond to BDS establishments (Section 2). Participation and its persistence discipline mainly the two export costs; the size distribution and the employment shares of entrants and exiters discipline the productivity process and the entrants’ productivity gap; and the exit rate disciplines the operating cost and the exit probability, $\delta = 5.3$ percent. The entry cost is implied by free entry: it equals 5.9 years of the profit of an average operating firm, net of its operating fixed cost.

The model column of Table 2 reports the balanced growth path at these parameters. It matches the six rate moments to within 0.14 percentage points. The size distribution is fit less well: the model places 60 percent of establishments in the smallest class, against 56 percent in the data, and 28 percent of employment in establishments with 100–499 employees, against 22 percent (Figure 2).

4.2 Demography and the Foreign Economy

Home labor-force growth is U.S. civilian labor-force growth. Foreign growth is export-weighted employment growth in the fifteen largest U.S. export markets (Appendix A). Both

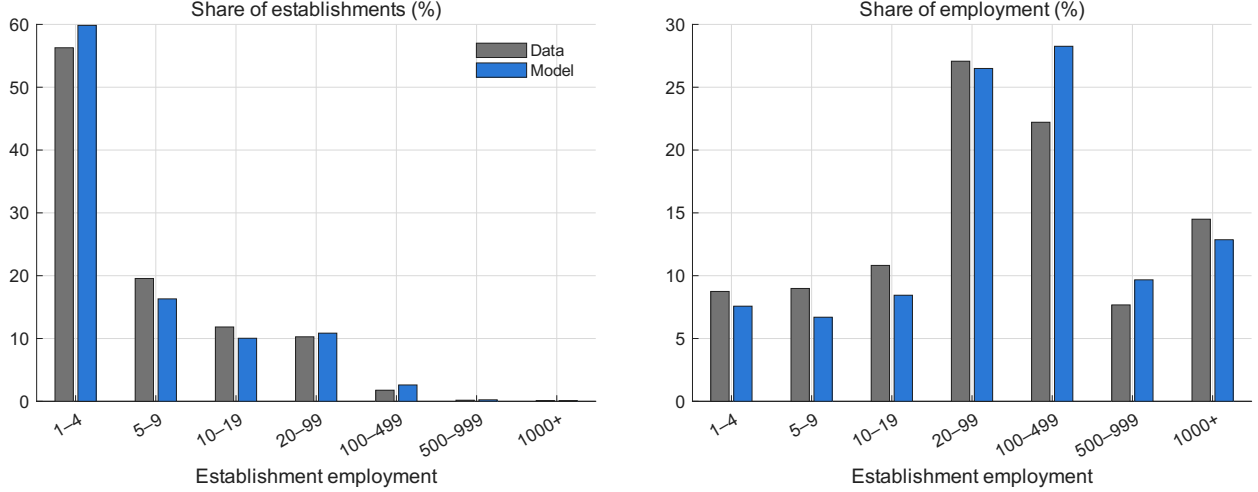


Figure 2: Establishment-size distribution, 1978: data and calibrated model. Shares of establishments and of employment by establishment employment.

are taken from the data through 2022, the last year of the accounting; the demographic inputs and the accounting window always end together. After 2022 both growth rates decline to zero along a cosine path over ten years and stay at zero, the common terminal growth that Proposition 1(i) requires. All results in the paper use this taper rather than published labor-force projections: in a perfect-foresight model, an anticipated reversal of the growth gap after 2022 would feed back into the fitted history. As a robustness check only, Section 6.6 replaces the taper with the projections; the demographic results do not change.

The foreign economy is the export-weighted rest of the world. It shares the home country's structural parameters, so the two economies differ only in their labor-force paths and in the wedges. This is a restriction, not an estimate: the foreign aggregate has no counterpart to the establishment data that discipline the home economy. Relative size is normalized to one in 1965; it is 0.93 in 1978 and 1.11 on the terminal balanced growth path. The foreign exporter share is not targeted; it is 12.0 percent in 1978, against 14.7 percent at home.

4.3 Accounting

Targets. Conditional on the structural parameters and the labor-force paths, the two iceberg wedges and the export-cost wedge are recovered in every year of 1965–2022 so that the model matches three series: home goods exports, the home exporter share, and the goods trade balance. Imports then follow from the external identity. The trade series are real goods exports and the real goods balance relative to real GDP (chained dollars). Because the final good is used only for consumption, we express them relative to domestic absorption, $Y - TB$, the data counterpart of consumption in the model. Entry, exit, and foreign participation

are not targeted. The exporter-share target before 1992 is constructed (Appendix A), so the early accounting partly inherits that construction.

Initial condition. The accounting starts from a symmetric balanced growth path at the 1965 labor-force growth rate, 1.71 percent in both countries. A common iceberg wedge of 1.38 and an export-cost wedge of 0.31 make this steady state reproduce 1965 home exports, 1.9 percent of absorption, and exporter participation, 14.0 percent, with balanced trade. The steady state supplies the initial firm distributions and a zero net foreign asset position. The wedge paths are revealed in 1965 and prices adjust on impact. The data pin only the product of the two iceberg wedges (Lemma 1); their split, and with it the level of the real exchange rate, is set by the closure of Section 4.4. In the baseline economy the recovered split therefore moves at once from the common value to 2.79 at home and 0.74 abroad, and the real exchange rate to 1.76. Section 6.6 reports what depends on the split.

Wedges at the calibration date. The calibration and the accounting target different export ratios. The cross-section targets the export intensity of exporters, a statistic about firms, and at wedges of one the calibrated steady state exports 6.7 percent of consumption, whereas U.S. real goods exports were 3.1 percent of absorption in 1978. The accounting reconciles the two with the wedges. In 1978 the geometric mean of the two iceberg wedges is 1.26, which lowers what each exporter sells abroad, and the export-cost wedge is 0.48, which keeps participation at its target. The levels of the wedges therefore absorb the gap between the firm-level and the aggregate export ratio, and the accounting in Section 6 works with their changes.

4.4 Closing the Path after 2022

A perfect-foresight path needs the exogenous inputs after the data end and a terminal condition for the external account. After 2022 the three wedges return, along a smooth path over thirty years, to values that describe a balanced growth path at zero labor-force growth with balanced trade, zero net foreign assets, and $q = 1$, and that reproduce the last observed home export ratio and exporter share. This follows Kehoe et al. (2018), whose fitted wedge reverts gradually after the data end. The premium is $\psi = 5 \times 10^{-4}$, and the path runs for 425 years with $B_{T+1} = 0$. The premium matters for the identification of the wedges: it sets the level of the real exchange rate and, with it, the split of the two iceberg wedges (Appendix D). It does not change the paper’s qualitative results, and Section 6.6 reports the demographic contributions under a premium four times larger. Appendix D explains each choice and reports the numerical criteria that every solution in the paper meets.

5 How Labor-Force Growth Reaches Exporters

The accounting of Section 6 recovers time-varying wedges that make the model reproduce the data. Before turning to it, we isolate the mechanisms in controlled transitions in which every wedge stays at its calibrated value and only labor-force growth moves. For these experiments we recalibrate the model to the same 1978 targets at a labor-force growth rate of one percent, the average trend growth of the U.S. labor force over 1985–2019 (Appendix F). In every experiment home labor-force growth stays close to one percent for a decade and then declines smoothly to zero, which it reaches after forty years.

5.1 The Symmetric Economy: Composition

In the first experiment both labor forces follow the same path. Trade stays balanced, the real exchange rate stays at one, and demography can reach exporters only through the composition of firms. Figure 3 shows the transition.

Slower labor-force growth does not change what any firm does. As growth falls, the interest rate falls, but the wage in which firms pay their operating, entry, and export costs grows more slowly in the same proportion, so the export and exit cutoffs are the same on the initial and the terminal balanced growth paths (Proposition 1). What changes is who the firms are. With fewer new workers each year, fewer new firms are needed, and incumbents carry more weight relative to entrants, as if firms survived longer (Corollary 1). Entrants start out as nonexporters and are less productive on average; incumbents have been selected by survival and have had time to grow into exporting. The firm population ages. The entry and exit rates fall, the average establishment grows, and the share of establishments that export rises, by 0.42 percentage points in the long run. Exports relative to consumption rise by less in proportional terms than participation, a first sign of the result in Section 6 that demography matters more for which firms export than for how much is traded.

The export decision adds this response without changing the response of business dynamism. Figure 3 also shows the same transition with both export costs switched off, the closed-economy limit of Section 3.9. Entry, exit, and average size follow almost the same paths in the two economies. Demography’s effect on business dynamism, the object of [Hopenhayn et al. \(2022\)](#) and [Karahana et al. \(2024\)](#), does not depend on whether firms export; what the open economy adds is the response of participation.

The composition channel is modest on the balanced growth path. Figure 4 compares balanced growth paths at the calibration of Section 4. Moving from the 1978 trend of labor-force growth, 2.64 percent, to the 2019 trend, 0.40 percent, raises the exporter share from 14.72 to 15.79 percent and lowers the entry rate from 14.1 to 11.0 percent. The long-run

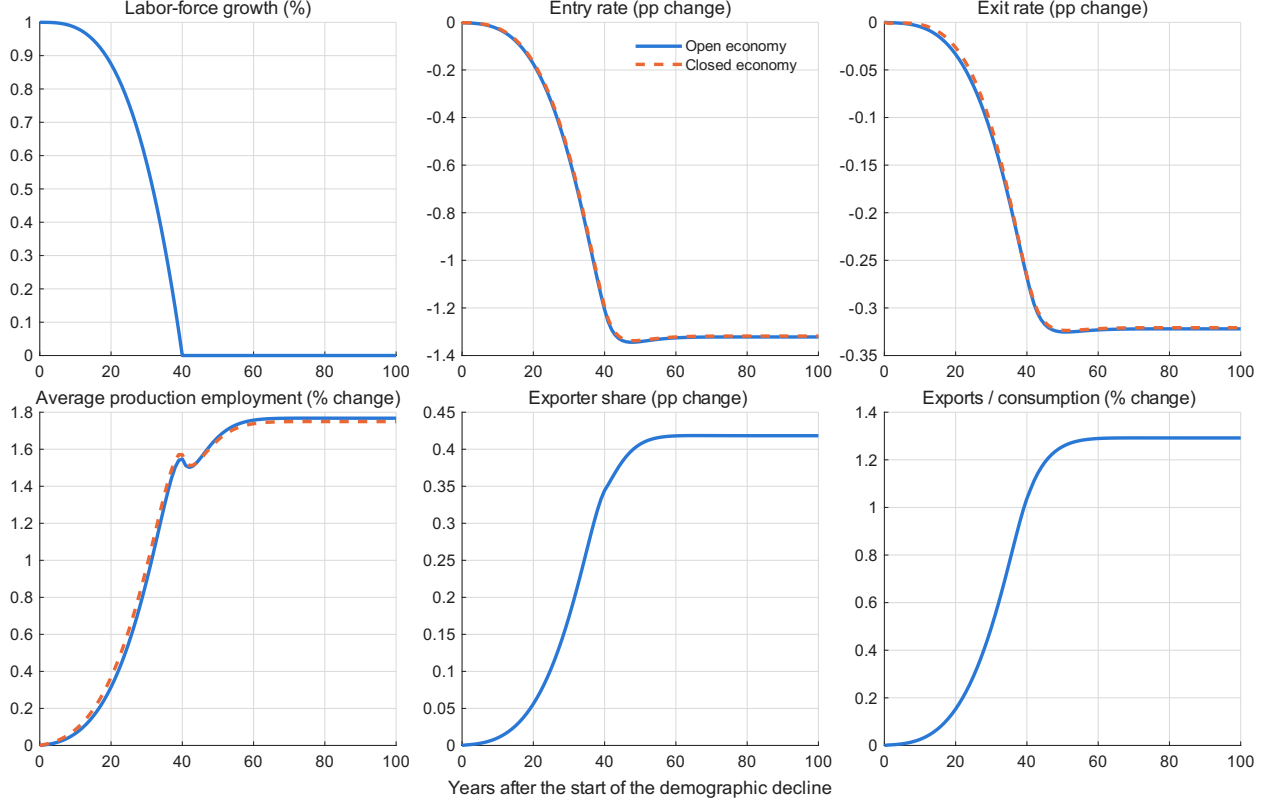


Figure 3: The symmetric controlled transition: labor-force growth falls from one percent to zero in both countries, every wedge at its calibrated value. Solid: open economy; dashed: closed-economy limit, both export costs set to zero at the same parameters. Entry rate: change from the initial balanced growth path; other panels: change from the first year. The exporter share and the export ratio are shown for the open economy only.

trade elasticity, $-d \log(X/C)/d \log(1+\tau)$ for a common cut in the gross iceberg cost, barely moves: it is 4.45 at the first growth rate and 4.41 at the second. Demography changes which firms export, not how strongly trade responds to trade costs. Studies of trade liberalization that hold labor-force growth fixed therefore miss demography's effect on participation, not an amplification of the trade response. The one-percent calibration, if anything, understates composition: at the structural parameters of the 1978 calibration, which attribute exporters' persistence to productivity rather than to a start-up premium, the same transition raises the exporter share by 0.59 percentage points rather than 0.42 (Appendix F).

5.2 The Asymmetric Economy: Relative Demand and Borrowing

In the other two experiments the home path is unchanged and foreign labor-force growth falls sooner and more slowly. Home therefore grows faster than foreign for thirty-five years and more slowly afterwards, the pattern of the United States relative to its trading partners

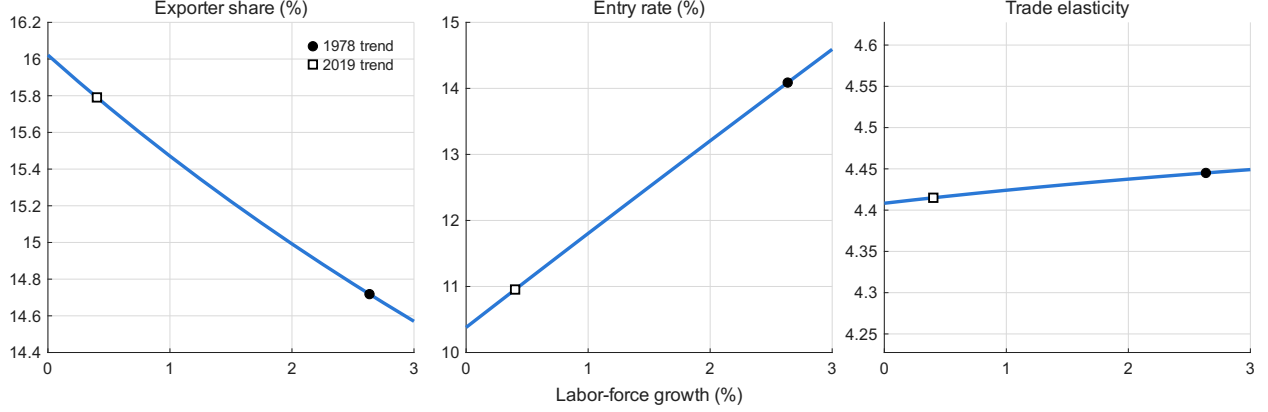


Figure 4: Balanced growth paths of the 1978 calibration at different rates of labor-force growth, every wedge equal to one. The trade elasticity is $-d\log(X/C)/d\log(1+\tau)$ for a one percent cut in the gross iceberg cost in both countries. Markers: the 1978 and 2019 trend growth rates of the U.S. labor force.

(Section 2); the foreign labor force shrinks to about 0.91 of the home one and then returns to parity. In one experiment the two countries cannot borrow from each other, so trade is balanced every period; in the other they trade the bond. Figure 5 shows both against the symmetric experiment. The entry and exit rates are the same in all three: the demographic gap acts on the export margin, not on business dynamism, because export profits are a small part of the value of a new firm.

Relative demand. With trade balanced, the gap reaches exporters through the relative profitability of exporting, which Proposition 2(i) splits into relative market size, the real exchange rate, and relative absorption. While home grows faster, its export market shrinks relative to its home market and exporting becomes less profitable. Relative prices cushion the fall. The home real exchange rate depreciates, and at the point of the largest effect it offsets 62 percent of the loss in relative market size. Participation falls below its symmetric path and catches up once relative size returns to parity.

Borrowing. With the bond, the timing of the gap matters. A faster-growing labor force devotes more of its workers to creating firms (Corollary 1(c)), so while home grows faster its income per worker is low relative to foreign, and it rises once home growth slows (Proposition 2(v)). Households borrow against that rise. Home runs trade deficits while it grows faster and surpluses once it grows more slowly, and its net foreign assets fall to -4.2 percent of consumption before the debt is repaid. Because iceberg costs bias each country's spending toward its own goods, borrowing raises the demand for home goods and strengthens the home real exchange rate, which makes exporting less profitable; repayment requires the op-

posite. Borrowing therefore first lowers home participation relative to balanced trade, by up to 0.13 percentage points, and then raises it, by up to 0.29. A larger debt premium makes households borrow less but leaves the pattern intact: with a premium four times larger, net foreign assets fall to -3.2 percent of consumption and borrowing moves participation between -0.11 and 0.25 points.

The three experiments separate the channels that the accounting combines. Composition operates in any economy and is the only channel in the symmetric one. Relative demand and borrowing exist only when labor forces grow at different rates, and they act through relative prices and the external account. The United States grew faster than its partners until the mid-1980s and more slowly afterwards, so over most of the accounting window it was in the second phase of both channels: relative demand recovering, and borrowing turned into lending. Both raise participation. Section 6 measures by how much.

6 Accounting for U.S. Exporter Participation, 1978–2022

The accounting recovers, year by year, the trade-cost and export-cost wedges with which the model reproduces U.S. exports, exporter participation, and the trade balance, given the observed labor-force paths (Section 4.3). With every wedge and both labor-force paths active, the model reproduces the three series exactly. We report the accounting over 1978–2022, the years after the calibration date; Appendix D shows the full paths, from 1965 to the end of the transition.

6.1 What the Accounting Recovers

Figure 6 shows the inputs. U.S. labor-force growth fell from about 2.6 percent a year in 1978 to below one percent by the 2010s; growth in U.S. export markets fell less, and from the mid-1980s the United States grew more slowly than its partners. Trade costs fell. The geometric mean of the two iceberg wedges, the trade-cost object the data identify (Lemma 1), falls from 1.26 in 1978 to 1.00 in 2022. How the fall is split between the two legs is set by the closure (Section 6.6), and we do not interpret it.

The export-cost wedge rises. In the model, falling trade costs alone would raise participation far more than it actually rose, and the accounting holds participation back with a rising cost of exporting. We read the wedge as the sum of everything that limited the extensive margin of exporting, not as a literal rise in the cost of starting to export, which would be implausible over a period of containerization, trade agreements, and the internet.

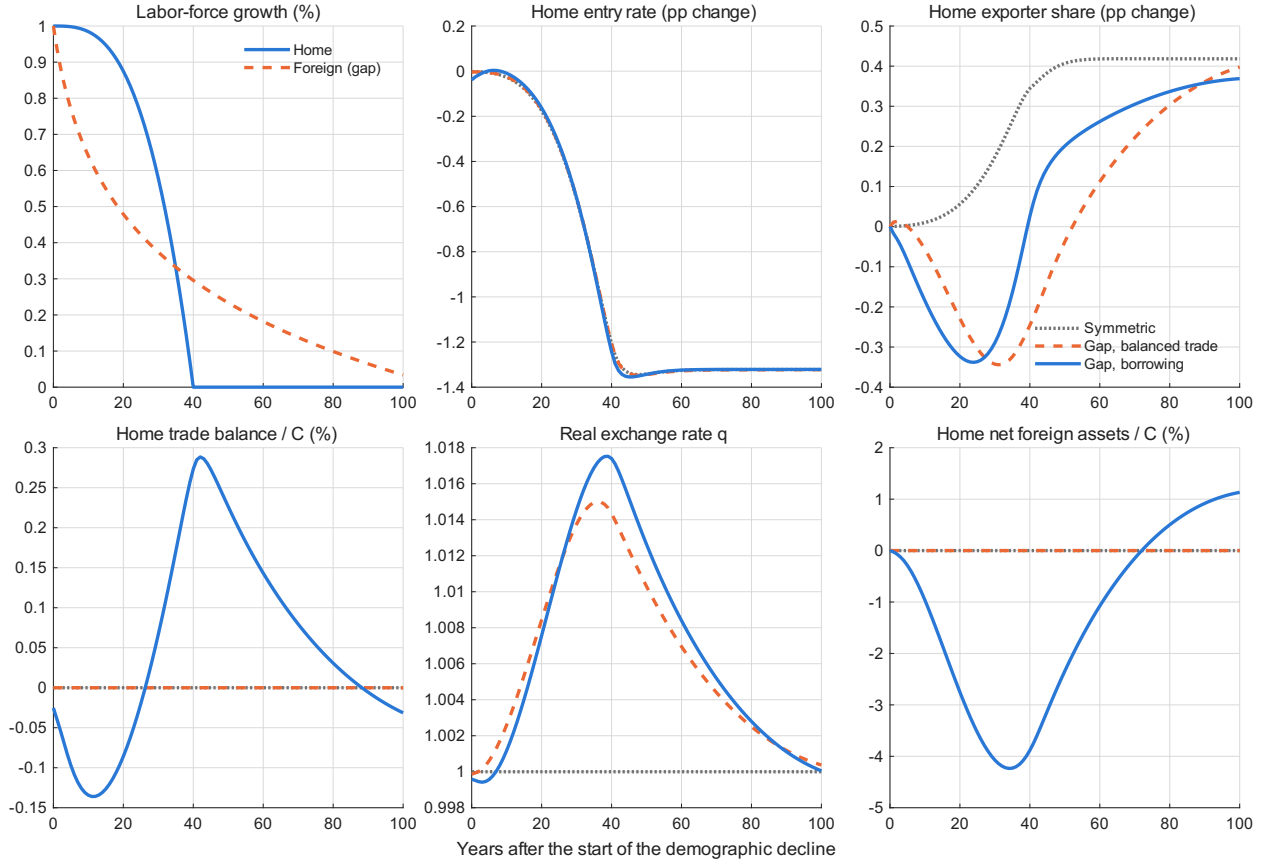


Figure 5: Controlled transitions with every wedge at its calibrated value. Top left: labor-force growth at home (solid) and abroad in the asymmetric experiments (dashed); in the symmetric experiment foreign growth equals home growth. Other panels: symmetric economy (dotted), asymmetric economy with balanced trade (dashed) and with borrowing (solid). Entry rate and exporter share are changes from the initial balanced growth path in percentage points; the trade balance and net foreign assets are percent of home consumption. Calibration at one percent labor-force growth (Appendix F).

Lincoln and McCallum (2018), studying the same period in Census microdata, likewise find that falling export entry costs do not explain the rise in exporting.

6.2 The Fitted Economy and the Labor Force

Figure 7 compares the data with the fitted economy and with a counterfactual in which only the labor force moves, every wedge held at its 1965 value. Labor-force growth alone produces trends: a steady decline in entry and exit, a gradual rise in average size, and a rise in participation of 2.0 percentage points over 1978–2022. It produces almost nothing in exports or the trade balance. Around these trends, the swings in participation and entry, the export boom of the 1990s and 2000s, and the trade deficits all come from the trade wedges.

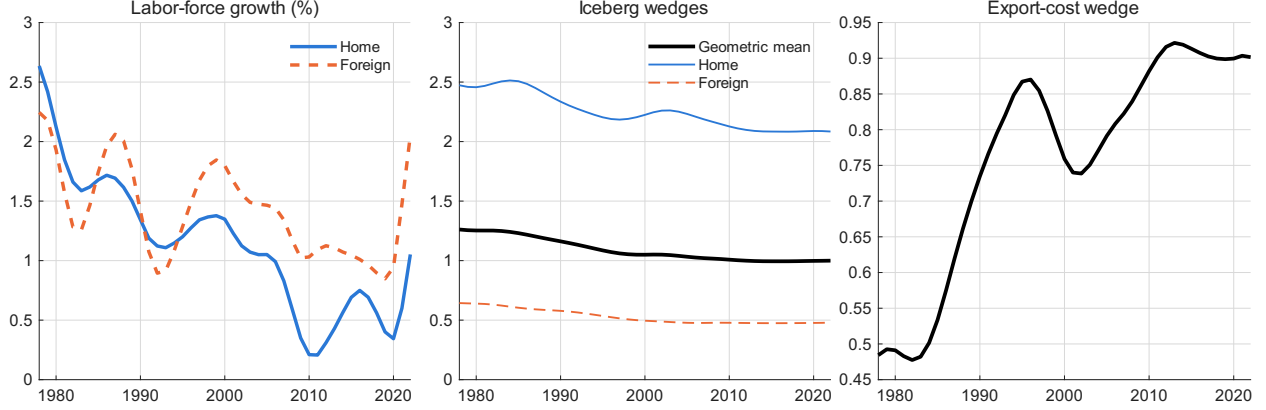


Figure 6: Inputs of the accounting, 1978–2022: labor-force growth at home and abroad, the recovered iceberg wedges (their geometric mean, which the data identify, and the home and foreign legs, whose split the closure sets), and the recovered export-cost wedge. Appendix D shows the paths from 1965 to the end of the transition.

Entry, exit, and size are not targeted. The fitted economy reproduces the direction of the decline in entry and its timing, and about half of its size: the model’s entry rate falls by 1.5 percentage points over 1978–2022, against 3.4 in the data. It misses most of the decline in exit and the rise in average size, which other forces must account for.

6.3 An Additive Decomposition

To say how much of each path a force accounts for, we decompose the fitted paths into three forces: trade costs (both iceberg wedges), the export-cost wedge, and labor-force growth. A force that is switched off is held at its 1965 value. We solve the model for each of the eight combinations of active forces and assign to each force its Shapley value, the average of its marginal contributions over the orders in which the forces can be switched on; the values sum to the outcome with every force active. Table 3 reports two statistics. The contribution to the path is the slope of the counterfactual on the observed path,

$$b = \frac{\text{Cov}(\Delta y_t^{cf}, \Delta y_t^{data})}{\text{Var}(\Delta y_t^{data})}, \quad (23)$$

where Δy_t is the change since 1978, and measures how much of the movement over the whole window a force reproduces. The contribution to the change is the counterfactual change between 1978 and 2022 in percent of the observed change. Forces can work against each other, so contributions can be negative and the others then exceed the total.

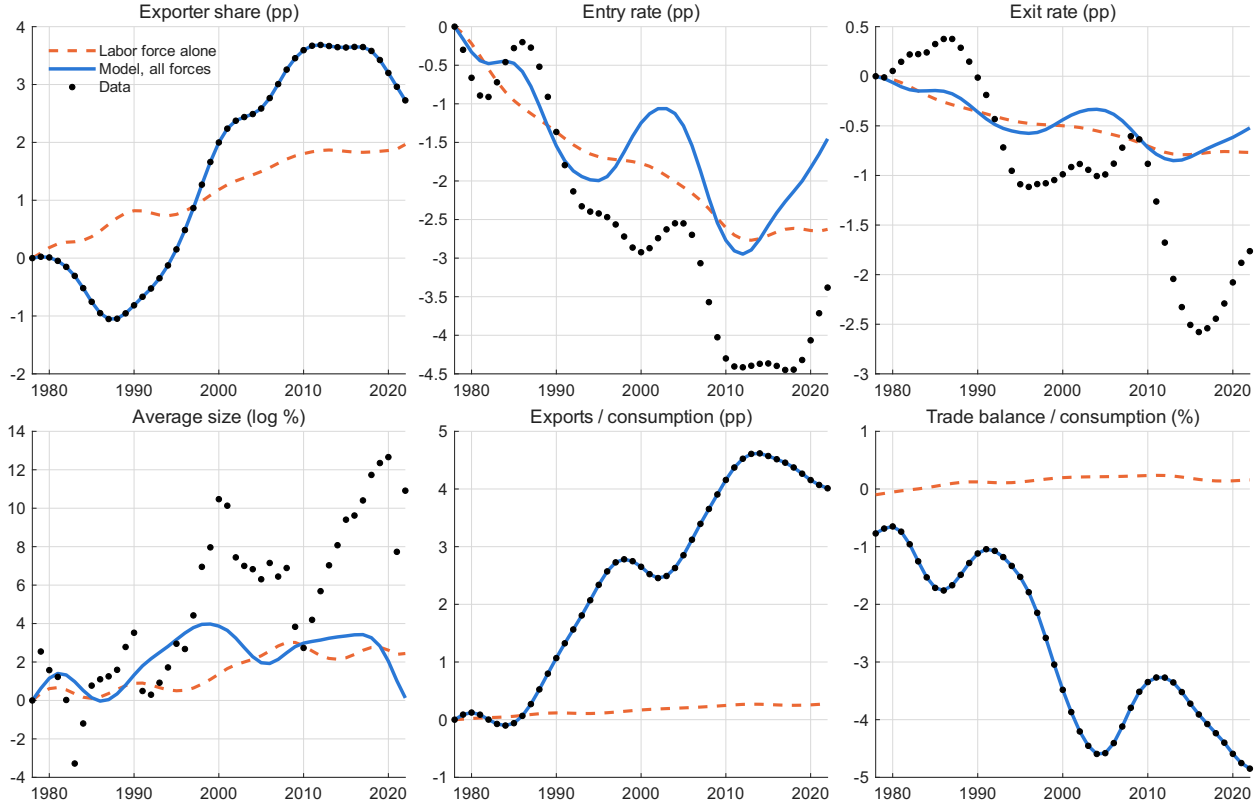


Figure 7: Data (markers), the fitted economy with every force active (solid), and labor-force growth alone with every wedge at its 1965 value (dashed), 1978–2022. Changes since 1978 except for the trade balance; average size in log percent. Exports, the exporter share, and the trade balance are targeted.

Demography carries business dynamism. On entry, exit, and average size, labor-force growth accounts for nearly everything the model produces, and the trade wedges for almost nothing. A slower labor force means fewer new firms and an older firm population, whatever happens to trade.

Demography is a first-order force behind which firms export. Labor-force growth accounts for about a third of the path of participation, 0.32. Trade costs and the export-cost wedge pull in opposite directions: falling trade costs push participation up by several times its observed rise, and the rising export-cost wedge holds most of that back. Demography’s contribution is not a by-product of that tug of war. Its Shapley value equals, to two decimals, its value when it acts alone, 0.32, so it does not depend on what the trade wedges do; grouping all trade frictions into a single force leaves it nearly unchanged (Appendix E).

Demography is a minor force behind how much is traded. Trade costs account for nearly all of the path of exports relative to consumption, and labor-force growth for 0.13.

The two statistics. For participation the contribution to the change, 73 percent, is much larger than the contribution to the path. Demography reproduces the long rise in participation but not the swings around it, which the path statistic counts against it; and the change statistic depends on where the window ends, since participation fell in the last years of the sample while demography kept raising it. Over 1992–2022, when the exporter share is directly observed, the contribution to the path is 0.32 and the contribution to the change 43 percent (Appendix E). We therefore use the contribution to the path as the main measure.

The demographic gap. Panel B repeats the decomposition in the symmetric economy, in which U.S. partners follow the U.S. labor-force path. Demography’s contribution to entry and exit is unchanged, but its contribution to participation falls to 0.13. The demographic effect on business dynamism is domestic; the effect on exporting depends on U.S. demography relative to its partners’. Section 6.5 decomposes the difference.

The iceberg split. The baseline evaluates demography at the iceberg split it recovers for 1965, which the closure sets (Section 4.3). The gap economy keeps the demographic gap but imposes a common iceberg wedge and leaves the trade balance untargeted, so it does not depend on the split. There labor-force growth alone accounts for 0.29 of the participation path, against 0.32 in the baseline. We keep the baseline as the headline because it is the only economy that also matches the trade balance, which disciplines borrowing; the gap economy shows that the demographic result does not rest on the split.

6.4 Entry: Trend and Swings

Figure 7 suggests a division of labor on entry, and Table 4 measures it. Each entry path is split into a trend and swings around it. Labor-force growth accounts for most of the trend of entry and almost none of its swings; trade costs account for none of the trend and for most of the swings, which they reproduce with a correlation of 0.77. In the data entry is above its trend when the U.S. trade deficit is above its own (correlation -0.74): periods of high domestic absorption are periods of high entry. The model reproduces this pattern, which it does not target, through the trade wedges, which also drive the trade balance and the real exchange rate. Demography sets the trend of entry; trade sets its swings.

6.5 Where Demography’s Effect on Participation Comes From

The controlled transitions of Section 5 separate three channels; four economies bring them in one at a time on the historical record. The symmetric economy has composition only. The

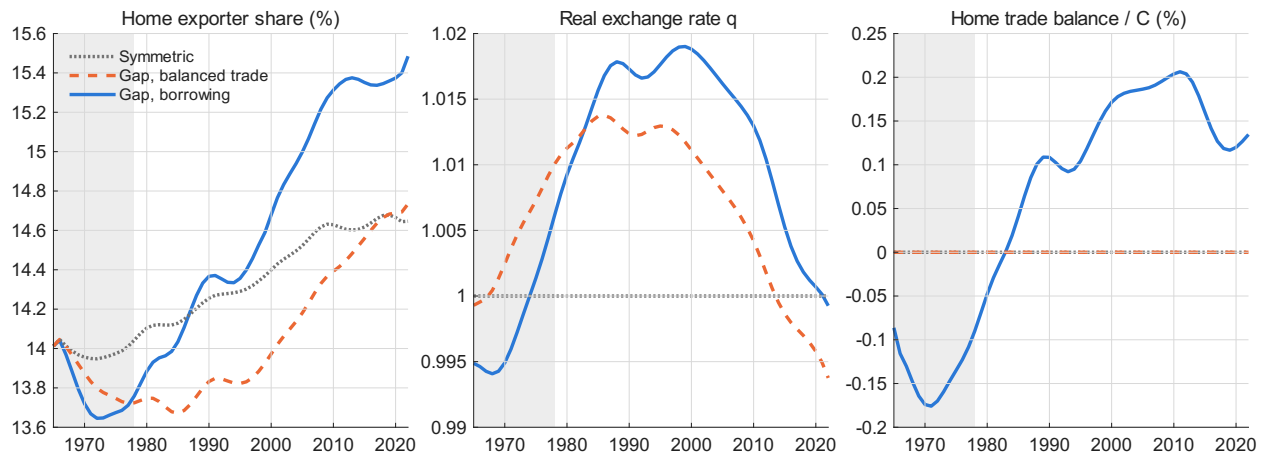


Figure 8: Labor-force growth alone, wedges held at their 1965 values, in the symmetric economy (dotted), the gap economy with balanced trade (dashed), and the gap economy with borrowing (solid), 1965–2022: home exporter share, real exchange rate (a rise is a home depreciation), and home trade balance relative to consumption. The shaded years precede the reporting window.

gap economy with balanced trade adds relative demand, the gap economy with borrowing adds the external account, and the baseline adds the split of the iceberg wedges. All four start from the same 1965 steady state, share the calibration, and are fed the observed labor-force paths. Of demography’s contribution to the path of participation in the baseline, composition accounts for 34 percent, relative demand for 23 percent, borrowing for 33 percent, and the iceberg split for 10 percent. More than half of demography’s effect on participation therefore comes from the demographic gap with trading partners, which a symmetric model cannot capture, and more than half of that runs through borrowing.

Figure 8 shows how. The U.S. labor force grew faster than its partners’ until the mid-1980s and more slowly afterwards. With borrowing, the United States borrows early and lends later, running small trade surpluses from the mid-1980s on. Running those surpluses requires a weaker real exchange rate than under balanced trade, by 0.8 percent on average over 2000–2010, and the weaker exchange rate makes exporting more profitable and draws marginal firms into it. The external flows involved are small; what matters for exporters is the relative price that comes with them.

6.6 What Depends on the Closure

The demographic results do not depend on how the path is closed after 2022. Table 5 reports labor-force growth’s contribution to the paths of entry, participation, and exports under a shorter horizon, a debt premium four times larger, and the published labor-force projections

in place of the taper; none moves by more than three hundredths. What the closure does set is the split of the iceberg wedges, and with it the level of the real exchange rate. With the larger premium the ratio of the foreign to the home iceberg wedge in 2022 is 0.12 rather than 0.23. We therefore attach no interpretation to the separate wedges, report the four-force decomposition that separates them only in Appendix E, and rely on the gap economy, which has no split, to show that the demographic results do not depend on it.

7 Welfare

How much did the observed path of labor-force growth change welfare? We compare two solved equilibria that differ only in demography; every wedge stays at its 1965 value in both. In the first, the two labor forces follow their observed paths through 2022 and then the ten-year taper to zero growth of Section 4.2. In the comparison, both labor forces grow at their common 1965 rate, 1.71 percent a year, through 2022 and then follow the same taper. Both paths therefore share the same future, and the comparison measures the history of demography rather than an assumption about growth after 2022. The welfare effect λ_c is the permanent proportional change in per-capita consumption that makes a household in country c indifferent between the two paths,

$$\sum_t \beta^t \log c_{c,t}^{\text{obs}} = \sum_t \beta^t \log [(1 + \lambda_c) c_{c,t}^{\text{comp}}], \quad (24)$$

evaluated with the calibrated discount factor, $\beta = 0.966$, over the solved horizon. Table 6 reports it, and Figure 9 shows the paths behind it.

The history is not a simple slowdown. As the baby-boom cohorts entered the labor force, home labor-force growth rose well above its 1965 rate in the 1970s and fell below it only in the mid-1980s (Figure 9, upper left). By 1982 the home labor force was 11 percent larger than in the comparison; by 2022 it was 16 percent smaller, and after the taper the gap settles at 20 percent. The foreign labor force stayed closer to its 1965 rate and ends 10 percent smaller.

Two forces turn this history into welfare, and they pull in opposite directions. The first is the number of varieties. With love of variety at its standard value, per-capita consumption grows with the number of varieties, at the rate $G_L^{1/(\theta-1)}$ along a balanced growth path (Proposition 1(ii)). This is the scale effect of semi-endogenous growth (Jones, 1995, 2022): the level of per-capita consumption rises with the level of the labor force, with elasticity $1/(\theta-1) = 0.25$. The boom raised home per-capita consumption through this channel by up to 2.7 percent in 1982; the slowdown that followed leaves it permanently 4.9 percent lower.

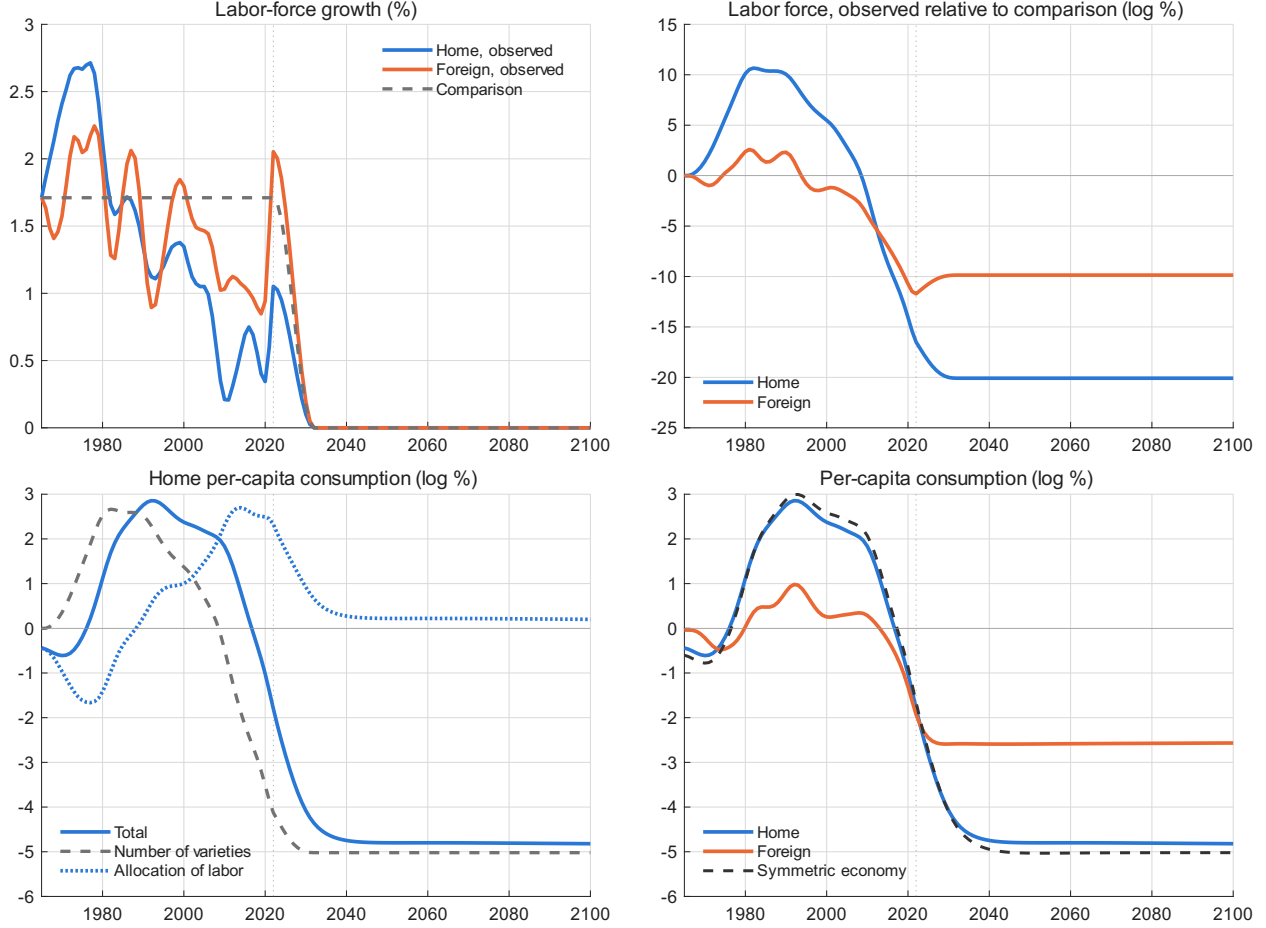


Figure 9: The observed labor-force path and the welfare comparison.

Notes: Baseline economy, every wedge at its 1965 value in both paths. The comparison holds labor-force growth at its 1965 rate through 2022 (dotted vertical line) and then applies the same taper as the observed path. Consumption and labor forces are log differences between the observed path and the comparison. In the lower-left panel, “number of varieties” is the difference in the per-capita growth trend, $(1+n_{H,t})^{1/(\theta-1)}$ cumulated, and “allocation of labor” is the difference in detrended consumption.

The second force is the allocation of labor. A labor force that grows faster needs more new firms to employ it, so more labor goes into creating firms and less into producing goods (Corollary 1(c)). Detrended home consumption, which moves with this allocation, was 1.7 percent lower than in the comparison in 1977, at the height of the boom, and 2.7 percent higher in 2014, after three decades of slower growth. Once growth is zero in both paths, the allocation of labor is almost the same in both and this force all but disappears.

The net effect on welfare is small. At home, the variety gains of the boom years and the allocation gains of the slow years arrive early and roughly offset the permanent scale loss, which arrives late and is discounted: the observed path is worth 0.18 percent of consumption. Abroad, where the labor force grew close to its 1965 rate until 1990, the early gains are much smaller and the welfare effect is -0.30 percent. Weighting each period by its population gives

more weight to the later, larger generations that bear the permanent loss, 0.02 percent at home and -0.43 percent abroad. In the symmetric economy the home effect is 0.16 percent: what matters for home welfare is the domestic labor-force path, not the demographic gap or the external adjustment it sets off. The long-run losses are not small: per-capita consumption ends 4.8 percent lower at home and 2.5 percent lower abroad. The welfare effect is small because they arrive late.

The comparison path matters for this conclusion. If labor forces kept growing at the 1965 rate forever, the observed path would cost 0.75 percent of consumption at home and 1.22 percent abroad, but nearly all of that loss comes after 2022, from growth the observed path never had; it measures the end of labor-force growth, not its history. Two further features matter for reading the numbers. First, the variety force rests on love of variety, which the calibration does not discipline against observed income growth. Under log utility the allocation of labor between creating firms and producing goods does not depend on the strength of love of variety (Appendix C); love of variety determines only how the number of varieties translates into consumption. Second, the wedges are held at their 1965 values in both paths, so the exercise measures the welfare effect of demography alone, not that of falling trade costs.

8 Regional Evidence

Of the three channels, only composition has a regional counterpart. Within the United States, places where the labor force grows more slowly should see less entry and, through the composition of firms, a larger share of establishments in exporting firms; relative demand and the external account act at the national level and are absorbed by year effects. We look for this pattern across the nine Census divisions, the finest geography at which the BDS trade supplement reports exporter shares. The supplement classifies an establishment by the trade status of its firm, so every establishment of a multi-unit exporter counts as exporting wherever it is located; the regional share therefore measures where exporting firms operate, not where exports originate. Following the demographic approach of [Shimer \(2001\)](#) and [Karahan et al. \(2024\)](#), we instrument labor-force growth with the state birth rate twenty years earlier, aggregated to divisions with population weights, and estimate

$$\text{ExporterShare}_{d,t} = \alpha + \gamma \text{LFgrowth}_{d,t} + \mu_d + \mu_t + \varepsilon_{d,t} \quad (25)$$

on division-year observations from 1992–2022, with standard errors clustered by year.

Table 7 reports the results. The pooled estimate is negative and significant, but it

reflects persistent differences across divisions: with division and year effects it is small and statistically insignificant, -0.04 with a standard error of 0.06 . The instrumental-variables estimate is negative and larger, -1.07 , but its first stage is weak, with an F statistic of 3.52 , so conventional standard errors are unreliable (Andrews et al., 2019; Lee et al., 2022). The exclusion restriction is also demanding: births twenty years earlier must affect the exporter share only through the labor force, not through local demand, schooling, or the sectoral composition of a division.

Two features of the design call for care with inference: the first stage is weak, and there are few clusters in either dimension, 31 years and nine divisions. Appendix Table 13 reports tests robust to each. The Anderson and Rubin (1949) test, which remains valid however weak the instrument, rejects $\gamma = 0$ when errors are clustered by year ($p < 0.001$) and with Driscoll and Kraay (1998) standard errors, which allow for correlation across divisions and over time ($p = 0.021$). The 95 percent Anderson–Rubin set clustered by year, $(-\infty, -0.45] \cup [9.63, \infty)$, excludes zero and contains the estimate in its negative part, but it is unbounded because the first stage is weak. Clustering by division allows for any serial correlation within a division but leaves only nine clusters, and it removes the rejection: the wild-bootstrap p -value is 0.141 , and the 95 percent set, $(-\infty, 0.55]$, contains zero. Wald tests by wild bootstrap (Davidson and MacKinnon, 2010; Roodman et al., 2019) give the same split, $p = 0.016$ with year clusters and $p = 0.153$ with division clusters. With a fifteen-year lag of the birth rate the set clustered by year is bounded and negative, $[-4.01, -0.18]$; with a twenty-five-year lag the point estimate changes sign (Appendix H).

Averaging over periods removes year-to-year noise and turns the regression into its long-difference version. With five-year periods, ten-year periods, or the two halves of the sample, and division and period effects, the instrumental-variables estimate ranges from -1.68 to -1.01 , close to the annual estimate. With only nine division clusters the evidence is weak: the wild-bootstrap p -values of the Wald test are 0.220 or higher, and those of the Anderson–Rubin test range from 0.048 to 0.157 (Appendix Table 14). Without the instrument the estimates are small and negative, between -0.12 and -0.11 .

The entry margin is clearer. Entry rises with labor-force growth in every specification with the twenty-year lag. In the annual regression the Anderson–Rubin set clustered by year is $[0.45, \infty)$, which excludes zero, and with period averages the instrumental-variables estimates range from 0.85 to 1.17 , with wild-bootstrap p -values between 0.055 and 0.137 . Instrumenting the entry rate itself with the lagged birth rate gives a strong first stage, $F = 32.30$, and a coefficient on the exporter share of -1.12 , with a bounded Anderson–Rubin set, $[-1.59, -0.66]$, clustered by year; its exclusion restriction requires past births to affect participation only through entry. The signs are those of the composition channel. Across

the balanced growth paths of Figure 4, one percentage point of labor-force growth raises the entry rate by 1.40 points and lowers the exporter share by 0.48 points: the regional estimates are close to the first number and larger than the second. Regional differences are not differences between balanced growth paths, however, since divisions trade with one another and workers move between them, so we do not read the estimates as a test of these magnitudes.

Taken together, the regional evidence agrees with the model in sign on both margins, more clearly for entry than for participation. Both relations are significant when the errors may be correlated across divisions within a year, and neither is significant at the 5 percent level when the errors may be serially correlated within a division and only nine clusters remain. With nine divisions and three decades of data the design has too little identifying variation to settle the question, and the quantitative case rests on the model and the accounting.

9 Conclusion

Over the past four decades the United States has created fewer firms, a larger share of its establishments has come to belong to exporting firms, and its labor force has come to grow more slowly than those of its trading partners. This paper argues that these changes are connected, and that the connection runs through two channels that leave different footprints.

The first channel is the composition of firms. In a symmetric world with log utility and with entry and export costs paid in labor, labor-force growth changes no firm’s decision to start exporting, to stop exporting, or to exit along a balanced growth path: the invariance of incumbents’ decisions known from closed economies extends to the sunk-cost export decision. A slower labor force therefore raises participation only by tilting the population of firms from entrants, which start as nonexporters, toward incumbents that have already paid to export. The second channel exists only because labor forces grow at different rates. Relative market size and international borrowing then move the real exchange rate, and with it the profitability of exporting. Entry responds to a country’s own labor force; exporter participation responds also to its labor force relative to its partners’.

The two margins respond to demography very differently. In an accounting of 1978–2022 in which the model reproduces U.S. exports, exporter participation, and the trade balance, labor-force growth accounts for about half of the path of entry (0.52), about a third of the path of participation (0.32), and little of the path of exports (0.13). Its effect on entry is domestic and works on the trend: demography accounts for 55 percent of the trend of entry and almost none of its swings, which move with the trade balance. Its effect on participation is mostly international: composition, the only part a closed or symmetric

model can capture, accounts for 34 percent of it, and borrowing and the real exchange rate for 33 percent. Demography is thus a first-order but not a dominant force behind which firms export, and a minor force behind how much is traded; falling trade costs and a rising export-cost wedge account for most of the rest.

Two further results qualify these magnitudes. The welfare effect of the observed labor-force path is small, 0.18 percent of consumption at home and -0.30 percent abroad, although per-capita consumption at home ends 4.8 percent lower in the long run: the U.S. labor force grew faster than its 1965 rate before it grew more slowly, and the gains of the early years offset a loss that arrives late. Across Census divisions, the regional data agree with the model in sign, more clearly for entry than for participation, but they have too little identifying variation to test its magnitudes.

The results bear on how trade and business dynamism are studied. Accounts of trade liberalization that hold labor-force growth fixed attribute to trade part of a rise in participation that comes from an older population of firms. Demography changes which firms export, not how strongly trade responds to trade costs: the long-run trade elasticity barely moves with the growth rate of the labor force, 4.45 against 4.41. Closed-economy accounts of declining business dynamism are right about entry but miss most of its export counterpart, because that counterpart depends on relative demography. The same worldwide slowdown raises participation more in a country whose labor force slows more than its partners', and less in one whose labor force slows less.

Several questions remain open. Besides trade costs, the force that matters most for participation in the accounting is the export-cost wedge, which rises and holds participation back while trade costs fall; the accounting does not say what it captures. The foreign economy shares the home economy's structural parameters, a restriction that firm-level data for U.S. partners would relax. Log utility removes an interest-rate channel through which slower growth would change the option value of the sunk export cost, and infinitely lived households remove the life-cycle saving through which demographic change reshapes external balances in [Sposi \(2022\)](#) and [Nie et al. \(2024\)](#); either would add to the borrowing channel we identify. Finally, the model accounts for about half of the decline in entry and little of the decline in exit, so the forces that complete the account of declining business dynamism may matter for exporting as well.

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Table 2: Parameters and targeted moments.

<i>A. Parameters</i>		
Parameter	Symbol	Value
<i>Assigned</i>		
Elasticity of substitution	θ	5
Discount factor	β	0.966
Initial labor-force growth (%)	n_0	2.64
Scale of the operating shock	σ_x	0.001
Structural iceberg cost	τ	0.78
<i>Calibrated jointly</i>		
Start-up export cost	κ_0	0.0176
Continuation export cost	κ_1	0.0177
Productivity persistence	ρ_z	0.972
Productivity innovation s.d.	σ_z	0.126
Entrant productivity gap	μ_E	0.078
Operating fixed cost	f_O	0.0270
Exogenous exit probability	δ	0.053
<i>Implied</i>		
Entry cost	f_E	1.12
<i>B. Targeted moments, 1978 (percent)</i>		
Moment	Data	Model
Exporter share	14.72	14.72
Stopper ratio	17.00	17.00
Entry rate	14.20	14.09
Exit rate	11.31	11.45
Employment share of entrants	6.80	6.88
Employment share of exiters	5.27	5.20
Export intensity of exporters	9.03	9.03
Size distribution	Figure 2	

Notes: Costs and σ_x are in units of labor; an average operating firm employs 0.87 units in production. The model column is the balanced growth path at n_0 with every wedge equal to one, solved at the calibrated parameters. Data are for 1978 from the BDS, except the exporter share, constructed as described in Appendix A; the stopper ratio, from [Bernard and Jensen \(1999\)](#); and the export intensity, interpolated from [Bernard and Jensen \(1995\)](#). The two export costs are nearly equal ($\kappa_0/\kappa_1 = 0.99$) because, given the productivity process, productivity shocks alone generate the observed stopping: with every other parameter fixed, the stopper ratio is 16.9 percent at $\kappa_0 = \kappa_1$ and 14.1 percent with a 15 percent start-up premium. Export costs remain sunk: the cost of exporting in $t + 1$ is paid in t , before productivity in $t + 1$ is known.

Table 3: Shapley decomposition of the accounting, 1978–2022.

	Contribution to the path				Contribution to the change (%)			
	Trade costs	Export cost	Labor force	All forces	Trade costs	Export cost	Labor force	All forces
<i>Panel A. Baseline economy</i>								
Exporter share	2.21 [2.87]	−1.53 [−0.70]	0.32 [0.32]	1.00	372	−345	73	100
Entry rate	−0.04 [−0.04]	0.00 [0.00]	0.52 [0.52]	0.48	−36	0	78	43
Exit rate	−0.01 [−0.01]	0.00 [0.00]	0.23 [0.23]	0.23	−14	0	43	30
Average size	−0.05 [−0.09]	0.01 [0.00]	0.21 [0.18]	0.17	−25	1	24	1
Exports / consumption	0.99 [0.96]	−0.12 [−0.06]	0.13 [0.05]	1.00	97	−14	17	100
Trade balance / consumption	1.20 [1.35]	−0.02 [−0.01]	−0.18 [−0.05]	1.00	124	−3	−22	100
<i>Panel B. Symmetric economy</i>								
Exporter share	2.30 [3.00]	−1.43 [−0.64]	0.13 [0.11]	1.00	409	−332	24	100
Entry rate	0.00 [0.00]	0.00 [0.00]	0.52 [0.52]	0.52	−1	1	77	78
Exit rate	0.00 [0.00]	0.00 [0.00]	0.23 [0.23]	0.23	−1	0	43	42
Average size	0.07 [0.06]	0.01 [0.00]	0.17 [0.16]	0.25	8	2	20	30
Exports / consumption	1.08 [1.12]	−0.10 [−0.06]	0.02 [0.01]	1.00	110	−12	2	100

Notes: “Contribution to the path” is (23) over 1978–2022, with the value of the force acting alone in brackets. “Contribution to the change” is the 1978–2022 change in percent of the observed change. Forces that are switched off are held at their 1965 values. The contributions of each row sum to the total. In the symmetric economy the trade balance is zero throughout.

Table 4: The trend of entry and the swings around it, 1978–2022.

Counterfactual	Trend	Swings around trend	
	loading	loading	correlation
Symmetric economy: labor force only	0.55	0.06	0.34
Symmetric economy: all forces	0.55	0.13	0.56
Baseline economy: labor force only	0.55	0.05	0.28
Baseline economy: trade costs only	−0.10	0.52	0.77
Baseline economy: all forces	0.46	0.55	0.71

Notes: Each entry path is split into a Hodrick–Prescott trend ($\lambda = 100$) and the swings around it. “Loading” is the slope of the model component on the corresponding component of the data; “correlation” is the correlation of the model and data swings. Entry is not targeted.

Table 5: Demographic contributions across closures of the path after 2022.

Closure	ψ	T	Post-2022 demography	Labor-force growth alone: loading		
				Entry	Exporter share	Exports
Baseline	5×10^{-4}	425	taper	0.52	0.32	0.05
Shorter horizon	5×10^{-4}	250	taper	0.52	0.32	0.05
Larger premium	2×10^{-3}	250	taper	0.52	0.29	0.05
Projected demography	5×10^{-4}	425	projected	0.51	0.31	0.05

Notes: Each entry is the contribution of labor-force growth alone, wedges held at their 1965 values, to the path of the outcome over 1978–2022. “Taper” is the ten-year cosine decline of labor-force growth to zero after 2022 (Section 4.2); “projected” uses the published projections instead.

Table 6: Welfare effect of the observed path of labor-force growth, relative to growth at its 1965 rate through 2022 (percent of per-capita consumption).

	Baseline economy		Symmetric economy
	Home	Foreign	
Welfare effect, λ	0.18	−0.30	0.16
Population-weighted	0.02	−0.43	0.03
Long-run per-capita consumption	−4.80	−2.49	−4.90

Notes: λ in (24). Both paths are solved equilibria with every wedge at its 1965 value, and both follow the same ten-year taper to zero labor-force growth after 2022. “Population-weighted” weights each period by $\beta^t L_{c,t}$ along the observed path. “Long-run per-capita consumption” is the percent difference between the two paths once labor-force growth is zero in both. In the symmetric economy the two countries are identical.

Table 7: Exporter share and labor-force growth across Census divisions.

	(1) OLS	(2) OLS	(3) IV
Labor-force growth	−0.36*** (0.10)	−0.04 (0.06)	−1.07* (0.55)
Division and year fixed effects	No	Yes	Yes
First-stage F (Kleibergen–Paap)	—	—	3.52
Observations	279	279	279

Notes: Division-year observations from 1992–2022. The instrument in column (3) is the state birth rate twenty years earlier, aggregated to divisions with population weights. Standard errors in parentheses, clustered by year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, conventional inference.

A Data Sources and Construction

Aggregate series. The aggregate series run through 2022. Every series that the model targets or takes as an input is trend-filtered with a Hodrick–Prescott filter with annual parameter $\lambda = 6.25$. Home labor-force growth is the growth of the civilian labor force aged 16 and over, from the BLS via FRED ([U.S. Bureau of Labor Statistics, 2026](#)). Real GDP and real goods exports and imports are from the BEA national accounts, in chained dollars ([U.S. Bureau of Economic Analysis, 2026](#)); the accounting targets real goods exports and the real goods balance relative to real absorption, GDP minus the goods balance (Section 4.3). Foreign labor-force growth is export-weighted employment growth in the fifteen largest U.S. export markets ([World Bank, 2026](#)), combined with partner growth from the Penn World Table ([Feenstra et al., 2015](#); [Groningen Growth and Development Centre, 2025](#)) and a world-growth fallback from the United Nations ([United Nations Population Division, 2026](#)).

The demographic tail. Both labor-force series enter the model through 2022, the last year of the accounting. After 2022 each declines to zero along

$$n_{c,2022+s} = \frac{1}{2} \left(1 + \cos(\pi s/10) \right) n_{c,2022}, \quad s = 1, \dots, 10,$$

and is zero thereafter. The path is continuous at the end of the data and reaches zero with zero slope. We do not use published labor-force projections in the main results. The projected series is rounded to a tenth of a percentage point, and it reverses the sign of the foreign-minus-home growth gap immediately after 2022 and keeps it reversed; in a perfect-foresight model that reversal would feed back into the fitted history. Section 6.6 reports the demographic contributions under the projections.

Business entry, exit, and exporter shares. From 1978 on, establishment entry and exit rates are from the BDS ([U.S. Census Bureau, 2025](#)). The entry series before 1978 combines the early startup-rate history of [Hopenhayn et al. \(2022\)](#), the BDS-anchored imputation of [Karahan et al. \(2024\)](#) for 1966–1977, and a bridge for 1963–1965 from historical County Business Patterns counts ([Eckert et al., 2022](#)). Exporter shares come from the BDS trade-status tabulations, in which an establishment belongs to an exporting firm if its firm is a goods exporter or a goods importer-exporter; the tabulations begin in 1992. Earlier years are filled by a local-level Kalman filter and smoother in logit space that never uses contemporaneous trade flows. The state is the log-odds of the exporter share, and the measurement equations are the BDS itself from 1992 on, the exporter-share series of [Alessandria and Choi \(2019\)](#) from 1980 on, through an estimated affine measurement equation, and two Census

benchmarks, the plant-level export shares of [Bernard and Jensen \(1995, 1999\)](#) and the share of Census manufacturing plants with 20 or more employees that export reported by [Lincoln and McCallum \(2018\)](#), each with its own estimated level bias. Observed BDS values are never overwritten.

Composition decomposition. Table 1 uses the BDS goods-trader tabulations by firm age (0, 1–5, 6–10, and 11 or more years) and by firm size (1–19, 20–499, and 500 or more employees). Firms born before 1977, whose exact age the BDS does not report, are at least fifteen years old in 1992 and join the oldest group. With establishment shares w_g and exporter shares p_g , the change in the aggregate share splits exactly into $\sum_g \bar{p}_g \Delta w_g$ and $\sum_g \bar{w}_g \Delta p_g$, with bars denoting averages of the two years.

State and division panel. The regional panel is built at the state-year level and aggregated to the nine Census divisions, with the District of Columbia in the South Atlantic division. State labor force, population, and gross state product are from FRED, and establishment entry and exit from the BDS state tables ([U.S. Census Bureau, 2025](#)). Exporter shares are from the BDS trade-status tabulations by Census division, which start in 1992. The instrument is the state birth rate, births per thousand residents, lagged fifteen, twenty, or twenty-five years and aggregated to divisions with population weights; for 1995–2004 the series is extended with state birth counts relative to state population, spliced to the earlier series in 1995.

B Proofs and Additional Derivations

Notation and equilibrium conditions. Write $d_{c,t} \equiv (1 + \tau)\omega_{c,t}^\tau$ and $\vartheta \equiv \theta/(\theta - 1)$ for the markup. A country- c firm with state (z, e) that operates uses $l_{c,t}(z, e)$ units of labor in production, f_O in operating costs, and $\omega_t^\kappa \kappa_e$ in export costs if it chooses $e' = 1$. The labor shares of Section 3.5 are

$$\begin{aligned} \ell_{c,t}^P &= \frac{1}{L_{c,t}} \sum_e \int l_{c,t} \tilde{\mu}_{c,t} dz, & \ell_{c,t}^O &= \frac{f_O}{L_{c,t}} \sum_e \int \tilde{\mu}_{c,t} dz, \\ \ell_{c,t}^X &= \frac{\omega_t^\kappa}{L_{c,t}} \sum_e \int \kappa_e \mathbf{1}\{x_{c,t} = 1\} \tilde{\mu}_{c,t} dz. \end{aligned}$$

With markup pricing, the CES indices of the prices, in home goods, of the varieties sold in the two markets are

$$\tilde{P}_{H,t}^{1-\theta} = \vartheta^{1-\theta} [W_{H,t}^{1-\theta} \mathcal{Z}_{H,t} + (q_t d_{F,t} W_{F,t})^{1-\theta} \mathcal{Z}_{F,t}^x], \quad (26)$$

$$\tilde{P}_{F,t}^{1-\theta} = \vartheta^{1-\theta} [(q_t W_{F,t})^{1-\theta} \mathcal{Z}_{F,t} + (d_{H,t} W_{H,t})^{1-\theta} \mathcal{Z}_{H,t}^x]. \quad (27)$$

The aggregator (7) gives the final-good prices $N_{H,t}^{1/(\theta-1)-\nu} \tilde{P}_{H,t} = 1$ and $N_{F,t}^{1/(\theta-1)-\nu} \tilde{P}_{F,t} = q_t$, and demand $y_i = N_{d,t}^{(\theta-1)\nu-1} (p_i/P_{d,t})^{-\theta} D_{d,t}$ for a variety with price p_i in market d . Combining the two, the share of spending in market d that goes to that variety is

$$\frac{p_i y_i}{P_{d,t} D_{d,t}} = \left(\frac{p_i}{\tilde{P}_{d,t}} \right)^{1-\theta}, \quad (28)$$

for every ν . Writing $C_{c,t} = L_{c,t} c_{c,t}$ for aggregate consumption in the country's own good, the profit shifters are therefore

$$A_{HH,t} = \frac{1}{\theta} (\vartheta W_{H,t})^{1-\theta} \tilde{P}_{H,t}^{\theta-1} C_{H,t}, \quad A_{HF,t} = \frac{1}{\theta} (\vartheta d_{H,t} W_{H,t})^{1-\theta} \tilde{P}_{F,t}^{\theta-1} q_t C_{F,t}, \quad (29)$$

in home goods, and symmetrically for foreign firms. At $\nu = 1/(\theta-1)$, $\tilde{P}_{H,t} = 1$ and $\tilde{P}_{F,t} = q_t$, and (29) reduces to (9). An equilibrium satisfies, in each country and period: the household Euler equation $Q_{c,t} = \beta c_{c,t}/c_{c,t+1}$; the international condition (5) (or $TB_t = 0$ under balanced trade); the budget constraints and (6); the firm problem (10)–(12); free entry (13); the laws of motion (14); labor-market clearing (15); the two final-good price conditions, which with (28) clear the markets for varieties; $D_{c,t} = L_{c,t} c_{c,t}$; and bond-market clearing.

The product of relative export profitabilities.

Lemma 1 (Product of relative export profitabilities). *At every date, on and off a balanced growth path and for every ν ,*

$$\Gamma_{H,t} \Gamma_{F,t} = (d_{H,t} d_{F,t})^{1-\theta}, \quad (30)$$

independently of wages, absorption, relative prices, and demography.

Proof. By (29),

$$\Gamma_{H,t} = d_{H,t}^{1-\theta} \left(\frac{\tilde{P}_{F,t}}{\tilde{P}_{H,t}} \right)^{\theta-1} \frac{q_t C_{F,t}}{C_{H,t}}, \quad \Gamma_{F,t} = d_{F,t}^{1-\theta} \left(\frac{\tilde{P}_{H,t}}{\tilde{P}_{F,t}} \right)^{\theta-1} \frac{C_{H,t}}{q_t C_{F,t}}.$$

Wages cancel within each ratio, and relative prices and relative absorption cancel in the product. $\square$

The ratio of the two relative export profitabilities is

$$\frac{\Gamma_{H,t}}{\Gamma_{F,t}} = \left(\frac{d_{H,t}}{d_{F,t}}\right)^{1-\theta} \mathcal{P}_t^{2(\theta-1)} \left(\frac{q_t C_{F,t}}{C_{H,t}}\right)^2, \quad \mathcal{P}_t \equiv \frac{\tilde{P}_{F,t}}{\tilde{P}_{H,t}}.$$

Observed participation and trade restrict $\Gamma_{H,t}$ and $\Gamma_{F,t}$; by (30) they restrict the product of the two iceberg wedges independently of relative prices and absorption, while the ratio of the wedges is determined only jointly with them. This is why Section 3.4 summarizes trade costs by the geometric mean $\bar{\omega}_t^\tau$.

Proof of Proposition 2. We prove the relations for every ν , with q_t in (i)–(ii) replaced by the relative variety price index $\mathcal{P}_t \equiv \tilde{P}_{F,t}/\tilde{P}_{H,t}$; at $\nu = 1/(\theta - 1)$, $\tilde{P}_{H,t} = 1$ and $\tilde{P}_{F,t} = q_t$, so $\mathcal{P}_t = q_t$. (i) In the expressions for $\Gamma_{H,t}$ and $\Gamma_{F,t}$ in the proof of Lemma 1, $q_t C_{F,t}/C_{H,t} = s_t q_t C_{F,t}/C_{H,t} = s_t/\rho_t^c$, so $\Gamma_{H,t} = d_{H,t}^{1-\theta} \mathcal{P}_t^{\theta-1} s_t/\rho_t^c$ and $\Gamma_{F,t} = d_{F,t}^{1-\theta} \mathcal{P}_t^{1-\theta} \rho_t^c/s_t$. (ii) Divide (27) by (26) and multiply the numerator and the denominator by $W_{H,t}^{\theta-1}$, using $(q_t W_{F,t}/W_{H,t})^{1-\theta} = (\rho_t^W)^{\theta-1}$; this gives (19) with $\mathcal{P}_t$ on the left. For the comparative statement in the text: scaling the foreign measures by λ at a given ρ_t^W changes $\log \mathcal{P}_t^{1-\theta}$ by $(1 - \lambda_{F,t}^{IM} - \lambda_{H,t}^{IM}) d \log \lambda$, where $\lambda_{c,t}^{IM}$ is country c 's import share of spending on varieties, by (28); this is positive when the two shares sum to less than one, and then $\mathcal{P}_t$ falls. (iii) Substitute $Q_{c,t} = \beta c_{c,t}/c_{c,t+1}$ into (5) and rearrange: $\rho_{t+1}^c/\rho_t^c = \exp[-\psi(\tilde{b}_{H,t+1} - \tilde{b}_{F,t+1})]$. Under balanced trade $TB_t = 0$, and part (iv) gives $\rho_t^c = \rho_t^y$. (iv) The foreign trade balance, in home goods, is $-TB_t$, so world absorption equals world GDP: $L_{H,t} c_{H,t} + q_t L_{F,t} c_{F,t} = L_{H,t} y_{H,t} + q_t L_{F,t} y_{F,t}$. Dividing by $L_{H,t}$ gives $c_{H,t}(1 + s_t/\rho_t^c) = y_{H,t}(1 + s_t/\rho_t^y)$, and $TB_t/(L_{H,t} c_{H,t}) = y_{H,t}/c_{H,t} - 1$ yields (21). (v) A firm's revenue in each destination is ϑ times its production wage bill, and GDP is the revenue of the country's firms, so $y_{c,t} = \vartheta W_{c,t} \ell_{c,t}^P$; the expression for $\ell_{c,t}^P$ is (15).

The range of relative consumption when $\psi = 0$. By (iii), $\rho_t^c = \rho^c$ is constant. Iterating (6) forward and imposing the no-Ponzi and transversality conditions gives $B_0 = -\sum_{t \geq 0} (\prod_{k < t} Q_{H,k}) TB_t$. With $B_0 = 0$ the discounted trade balances sum to zero, and by (iv) each has the sign of $\rho_t^y - \rho^c$. Unless $\rho_t^y = \rho^c$ at every date, the trade balance takes both signs, so $\inf_t \rho_t^y < \rho^c < \sup_t \rho_t^y$. $\square$

Proof of Proposition 1. Under $\nu = 1/(\theta - 1)$, $\tilde{P}_{H,t} = 1$ and $\tilde{P}_{F,t} = q_t$. (i) On a balanced growth path per-capita measures are constant, so $\mathcal{Z}_{c,t} = L_{c,t} \hat{\mathcal{Z}}_c$ and $\mathcal{Z}_{c,t}^x = L_{c,t} \hat{\mathcal{Z}}_c^x$ with constant $\hat{\mathcal{Z}}$. By (26) and (28), home's import share is $\lambda_{H,t} = d_F^{1-\theta} (\rho_t^W)^{\theta-1} \mathcal{Z}_{F,t}^x / [\mathcal{Z}_{H,t} + d_F^{1-\theta} (\rho_t^W)^{\theta-1} \mathcal{Z}_{F,t}^x]$, and with $\hat{\mathcal{Z}}_F^x > 0$ its constancy requires $(\rho_t^W)^{\theta-1} s_t$ to be constant. Home exports are $\lambda_F q_t C_{F,t}$ and imports $\lambda_H C_{H,t}$, so $TB_t/C_{H,t} = \lambda_F s_t/\rho_t^c - \lambda_H$, which is constant on the path (by (6), $TB_t/C_{H,t} = (B_t/C_{H,t})(Q_{H,t} C_{H,t+1}/C_{H,t} - 1)$ with $Q_{H,t} C_{H,t+1}/C_{H,t} =$

$\beta G_{L,H}$). Hence s_t/ρ_t^c is constant. Home GDP is $(1 - \lambda_H)C_{H,t} + \lambda_F q_t C_{F,t}$, so $y_{H,t}/c_{H,t} = 1 - \lambda_H + \lambda_F s_t/\rho_t^c$ is constant, and likewise $y_{F,t}/c_{F,t}$; hence $\rho_t^y \propto \rho_t^c \propto s_t$. Per-capita labor allocations are constant, so by Proposition 2(v), $\rho_t^y \propto \rho_t^W \propto s_t^{-1/(\theta-1)}$. Together, $s_t^{\theta/(\theta-1)}$ is constant, so s_t is constant and the two labor forces grow at the same rate. Finite lifetime utility requires $\beta G_L < 1$. (ii) With s_t and ρ_t^W constant, (26) with $\tilde{P}_{H,t} = 1$ gives $W_{H,t}^{\theta-1} = \vartheta^{1-\theta} L_{H,t} [\hat{Z}_H + d_F^{1-\theta} (\rho^W)^{\theta-1} s \hat{Z}_F^x]$, so $W_{H,t} \propto L_{H,t}^{1/(\theta-1)}$; the foreign condition gives $W_{F,t} \propto L_{F,t}^{1/(\theta-1)}$ in foreign goods, and $q_t = W_{H,t}/(\rho^W W_{F,t})$ is constant. Per-capita GDP $y_{c,t} = \vartheta W_{c,t} \ell_c^P$ grows at G_p , and so does $c_{c,t}$ because $y_{c,t}/c_{c,t}$ is constant. If $\psi > 0$, constancy of ρ_t^c in Proposition 2(iii) requires $\tilde{b}_H = \tilde{b}_F$; since $\tilde{b}_H = B/C_H$ and $\tilde{b}_F = -B/(qC_F)$ have opposite signs unless $B = 0$, net foreign assets are zero. (iii) Under log utility $Q_c = \beta/G_p$. Detrend values by a trend X_t growing at G_p ; then $\mathcal{Q} = QX_{t+1}/X_t = \beta$. Profits are proportional to the shifters, which grow with the wage, and all fixed, export, and entry costs are proportional to the wage, so the detrended Bellman equation is stationary. Detrending by aggregate output, which grows at $G_L G_p$, instead introduces G_L into the recursion. (iv) In detrended per-capita form, the price indices, demands, labor-market clearing, the Bellman equation, and free entry contain G_L only through $\mathcal{Q} = \beta$. The detrended household budget contains the term $\beta G_L \hat{b}_{c,t+1}$, which vanishes at zero net foreign assets. The laws of motion (14) carry the factor $1/G_L$. (v) Profits are linear in the two shifters, costs are linear in the wage, and the smoothed value (12) is homogeneous of degree one in $(V^{\text{stay}}, \hat{W})$. Hence $\hat{V}_c$ is homogeneous of degree one in $(\hat{A}_{cc}, \hat{A}_{cd}, \hat{W}_c)$, and $\hat{V}_c/\hat{W}_c$ and both policies depend on the aggregate state only through $\hat{a}_{cc}$, $\Gamma_c = \hat{A}_{cd}/\hat{A}_{cc}$, and $\mathcal{Q} = \beta$. Dividing (13) by $\hat{W}_c$ gives one G_L -free restriction per country, and (30) a third. Under symmetry $\hat{a}_{HH} = \hat{a}_{FF}$ and $\Gamma_H = \Gamma_F$, so (30) gives $\Gamma = d^{1-\theta}$, and free entry pins $\hat{a}_{HH}$ because $\hat{V}/\hat{W}$ is strictly increasing in it. The cutoffs depend only on $(\hat{a}_{cc}, \Gamma_c, \beta)$ and are therefore independent of G_L . $\square$

Proof of Corollary 1. By Proposition 1(v), the operating and export policies and $(\hat{a}, \Gamma)$ do not depend on G_L , and neither does the labor a firm uses: its production labor is $(\theta - 1)\hat{a}(1 + e\Gamma)e^{(\theta-1)z}$, and its operating and export costs are f_O and $\omega^\kappa \kappa_e$. With stationary policies the operator $\mathcal{T}$ of (14) is time-invariant, and the stationary solution of (14) is

$$\hat{\mu} = \frac{m}{G_L} \sum_{k \geq 0} \varrho^k \mathcal{T}^k \phi_E^0, \quad \varrho \equiv \frac{1 - \delta}{G_L} < 1,$$

where ϕ_E^0 places the entrant distribution on non-exporters. (a) The shape of $\hat{\mu}$ depends on G_L only through ϱ , and m scales it. (b) Let $S_k > 0$ be the total mass of $\mathcal{T}^k \phi_E^0$, with $S_0 = 1$. The entry rate is $m/\int \hat{\mu} = G_L/\sum_{k \geq 0} \varrho^k S_k$; the numerator rises with G_L and every term of the denominator falls. (c) Let $\lambda_k > 0$ be the labor used in production, operating costs, and

export costs by the firms that operate in the measure $\mathcal{T}^k \phi_E^0$, which does not depend on G_L . Then $\ell^P + \ell^O + \ell^X = m a(G_L)$ with $a(G_L) = G_L^{-1} \sum_{k \geq 0} \varrho^k \lambda_k$ strictly decreasing in G_L . By (15), $m = 1/(a(G_L) + f_E)$, so $f_E m = f_E/(a(G_L) + f_E)$ is strictly increasing in G_L . $\square$

C Love of Variety

The baseline sets $\nu = 1/(\theta - 1)$, at which a larger labor force raises per-capita income through variety. This appendix shows that under log utility ν plays no other role: it does not affect any firm decision or the external account. Let $e_{c,t} \equiv c_{c,t}/W_{c,t}$ denote per-capita consumption in units of the country's labor, and let $\mathcal{P}_t \equiv \tilde{P}_{F,t}/\tilde{P}_{H,t}$ be the relative variety price index of Appendix B, which equals q_t at the baseline ν .

Proposition 3 (Love of variety). *Assume log utility. Given β , the initial measures, the initial ratio of net foreign assets to consumption, and the paths of the labor forces and the wedges, the equilibrium paths of firm values relative to the wage, firm policies, the measures of firms, entry, the allocation of labor, $e_{H,t}$, $e_{F,t}$, the relative wage ρ_t^W , relative consumption ρ_t^c and income ρ_t^y , the relative variety price index $\mathcal{P}_t$, and the trade balance and net foreign assets relative to consumption do not depend on ν . The parameter ν affects only the price of each final good in units of labor, $N_{c,t}^{1/(\theta-1)-\nu} \tilde{P}_{c,t}/W_{c,t}$, and through it real consumption and the real exchange rate $q_t = (N_{F,t}/N_{H,t})^{1/(\theta-1)-\nu} \mathcal{P}_t$.*

Two properties deliver the result. A variety's share of spending in its market, (28), does not depend on ν , so neither do firms' revenues in units of labor. And under log utility the household's discount factor for payoffs in units of labor, $Q_{c,t} W_{c,t+1}/W_{c,t} = \beta e_{c,t}/e_{c,t+1}$, does not depend on how many final goods a unit of labor buys. Because all costs are paid in labor, every equilibrium condition can be written in units of labor without ν . The result therefore rests on log utility: with curvature $\sigma \neq 1$, the discount factor in units of labor becomes $\beta(e_{c,t}/e_{c,t+1})^\sigma (W_{c,t+1}/W_{c,t})^{1-\sigma}$, and ν matters through the growth of the wage in final goods.

Proof of Proposition 3. Measure every quantity of country c in units of its labor: $e_{c,t} = c_{c,t}/W_{c,t}$, and home net foreign assets $B_t^L \equiv B_t/W_{H,t}$. We show that the equilibrium conditions, written in these units, do not contain ν . (a) *Revenues*. By (28), a home firm's revenue relative to its wage is

$$\frac{R_{H,t}(z, e)}{W_{H,t}} = \vartheta^{1-\theta} e^{(\theta-1)z} \left[\left(\frac{\tilde{P}_{H,t}}{W_{H,t}} \right)^{\theta-1} L_{H,t} e_{H,t} + e d_{H,t}^{1-\theta} \left(\frac{\tilde{P}_{F,t}}{W_{H,t}} \right)^{\theta-1} \frac{L_{F,t} e_{F,t}}{\rho_t^W} \right],$$

and by (26)–(27), $(\tilde{P}_{H,t}/W_{H,t})^{1-\theta} = \vartheta^{1-\theta}[\mathcal{Z}_{H,t} + d_{F,t}^{1-\theta}(\rho_t^W)^{\theta-1}\mathcal{Z}_{F,t}^x]$ and $(\tilde{P}_{F,t}/W_{H,t})^{1-\theta} = \vartheta^{1-\theta}[(\rho_t^W)^{\theta-1}\mathcal{Z}_{F,t} + d_{H,t}^{1-\theta}\mathcal{Z}_{H,t}^x]$. None of these contains ν ; the same holds for foreign firms with ρ_t^W replaced by its inverse, and for production labor, which is revenue divided by ϑW .

(b) *Discounting*. Under log utility, $Q_{c,t}W_{c,t+1}/W_{c,t} = \beta e_{c,t}/e_{c,t+1}$, the price of a unit of next period's labor in units of this period's labor. Because profits are given by (a) and all fixed, export, and entry costs are paid in labor, firm values divided by the wage solve (10)–(13) with this discount factor, and the operating, export, and entry decisions depend only on (a) and on it.

(c) *International Euler condition*. Multiplying both sides of (5) by $W_{H,t+1}/W_{H,t}$ gives $\beta(e_{H,t}/e_{H,t+1})e^{-\psi\tilde{b}_{H,t+1}} = \beta(e_{F,t}/e_{F,t+1})(\rho_{t+1}^W/\rho_t^W)e^{-\psi\tilde{b}_{F,t+1}}$, with $\tilde{b}_{H,t+1} = B_{t+1}^L/(L_{H,t+1}e_{H,t+1})$ and $\tilde{b}_{F,t+1} = -B_{t+1}^L\rho_{t+1}^W/(L_{F,t+1}e_{F,t+1})$.

(d) *External identity*. Dividing (6) by $W_{H,t}$ gives $(Q_{H,t}W_{H,t+1}/W_{H,t})B_{t+1}^L = B_t^L + TB_t/W_{H,t}$, where exports and imports in units of home labor follow from (a).

(e) Labor-market clearing (15) and the laws of motion (14) involve only the policies, entry, and labor in production, operating costs, and export costs. Conditions (a)–(e) and the initial and terminal conditions determine the paths of the policies, the measures, entry, the labor allocation, $e_{H,t}$, $e_{F,t}$, ρ_t^W , and B_t^L without reference to ν . The remaining objects in the statement are functions of these: $TB_t/C_{H,t} = (TB_t/W_{H,t})/(L_{H,t}e_{H,t})$, $B_t/C_{H,t} = B_t^L/(L_{H,t}e_{H,t})$, $\rho_t^c = \rho_t^W e_{H,t}/e_{F,t}$, $\rho_t^y = \rho_t^W \ell_{H,t}^P/\ell_{F,t}^P$, and $\mathcal{P}_t$ by the general form of Proposition 2(ii) proved in Appendix B. The parameter ν enters only the final-good price in units of labor, $N_{c,t}^{1/(\theta-1)-\nu}\tilde{P}_{c,t}/W_{c,t}$, which converts $e_{c,t}$ into real consumption and determines $q_t = (N_{F,t}/N_{H,t})^{1/(\theta-1)-\nu}\mathcal{P}_t$. $\square$

Numerical check. We solve the three controlled transitions of Section 5 (symmetric, gap with balanced trade, and gap with borrowing) at $\nu = 0$, with the same β and all other parameters at their baseline values. Over the first 200 years, the paths of both countries' exporter shares, the entry and exit rates, entrants per worker, per-capita consumption in units of labor, the relative wage, and the trade balance and net foreign assets relative to consumption coincide with the baseline paths to within 10^{-8} , the tolerance of the solver. The objects that the proposition allows to differ do differ. Real per-capita consumption growth differs by up to 0.02 percentage points a year. In the gap transition with the bond, the real exchange rate rises to 1.018 at the baseline ν and falls to 0.994 at $\nu = 0$. It thus moves in opposite directions while the exporter paths are identical, so the real exchange rate by itself is not a sufficient statistic for the incentive to export.

D Transition, Closure, and Computation

Solving the transition. Values are solved backward and firm distributions forward, and prices adjust until the goods, labor, Euler, and external conditions hold. Net foreign assets are not carried as unknowns: given $B_{T+1} = 0$, the external identity (6) is solved backward for the whole path, which makes every annual budget equation hold exactly and leaves the observed initial position as one equilibrium condition. At a terminal balanced growth path the external identity implies $TB = (QG_L G_p - 1)B$, so the zero terminal position also implies a zero terminal trade balance. Under balanced trade, the T external conditions of the bond economy (the international Euler conditions and the initial-position condition) are replaced by $TB_t = 0$ in every period, the system remains square, and $B_t \equiv 0$.

Why the wedges must revert. The last recovered wedges and the terminal wedges are separate objects: the first are estimated from the data, the second solve the terminal balanced growth path. Two designs that dispense with a reversion both fail. Freezing the wedges at their 2022 values while keeping the terminal path leaves no equilibrium. Freezing the wedges and solving the terminal path at the frozen values is circular and diverges: without a terminal restriction nothing pins the split of the two iceberg wedges (Lemma 1), and the recovered home wedge runs off along that direction. The reversion therefore moves the economy to its terminal state and ties the tail to the terminal restriction that pins the split. It also schedules the adjustment after the sample: home entry falls 3.44 percentage points below its 2022 value, bottoming out in 2049, and net foreign assets reach -4.28 times consumption in 2045 as the external position inherited from the sample is unwound. These movements are closure dynamics, not recovered history, and the figures stop in 2022.

How fast the data say the wedges revert. A first-order autoregression of each wedge's in-sample log distance from its terminal value gives a half-life of about 106 years for the home iceberg wedge, a half-life of about 16 years for the export-cost wedge, and no reversion (the distance from its terminal value widens over the sample) for the foreign iceberg wedge. The thirty-year reversion closes half the distance in fifteen years. Symmetric long-run trade costs are therefore an assumption the sample does not reveal, as they are in [Kehoe et al. \(2018\)](#) and [Alessandria and Choi \(2021\)](#). Section 6 shows that the demographic results do not depend on the split between the two iceberg wedges that this assumption helps determine.

The premium and the horizon. The premium ψ is a stationarity device, but its level is not innocuous. Relative consumption is solved backward from the terminal date, so its in-window level, and with it the real exchange rate, inherits the premium accumulated over

the whole horizon, and through Lemma 1 that level lands in the split between the two iceberg wedges. Larger values settle the external position sooner but drive the recovered split toward extremes: at $\psi = 2 \times 10^{-3}$ the ratio of the foreign to the home iceberg wedge is 0.12 in 2022, against 0.23 at our value. Smaller values give a milder split but a slower settlement. We set $\psi = 5 \times 10^{-4}$ and a horizon of $T = 425$ years, long enough for the tail to settle; over the fitted window, lengthening the horizon from 250 to 425 years moves the recovered paths by at most 1.1 percent.

The terminal external position. The terminal condition $B_{T+1} = 0$ is a normalization; a nonzero terminal position would require a permanent nonzero terminal trade balance that a finite horizon cannot build up to.

Acceptance. We accept a solution only if it meets criteria stated in advance: the equilibrium residual, the distance to the matched targets (for the wedge recovery), and the settlement of the path, measured by the largest yearly drift of the firm moments and the largest one-year step of the aggregate levels over the last twenty years of the horizon. The wedge recovery solves the equilibrium conditions to within 1.6×10^{-5} and matches its targets to within 2.3×10^{-6} ; its terminal drift is 4.1×10^{-3} per year and its largest terminal step 1.8×10^{-3} . Re-solving the fitted economy independently, with the recovered wedges held fixed, reproduces its outcomes and levels to machine precision. The 24 counterfactuals of the decompositions (baseline and symmetric economies) solve to within 9.7×10^{-7} , with a largest terminal drift of 4.7×10^{-3} per year and a largest terminal step of 1.8×10^{-3} . The 4 demography counterfactuals of the nested specifications solve to within 3.0×10^{-7} (drift 1.7×10^{-4}), and the 6 controlled transitions of Section 5 and Appendix C to within 5.4×10^{-8} (drift 1.1×10^{-4}). Every one of these solutions is accepted.

The full path. Figures 10 and 11 show the inputs and the fitted economy from 1965 to the end of the transition. After 2022 the labor-force growth rates taper to zero and the wedges revert, over thirty years, to their common terminal values, and every path approaches the terminal balanced growth path. Because the recovered home iceberg wedge lies far above the foreign one, the reversion lowers the home trade cost sharply after 2022. The fitted economy responds with a transitory expansion of home exports, participation, and the trade balance that peaks around 2050 and then dies out. The counterfactual with labor-force growth alone has no such episode, because its wedges stay at their 1965 values. These paths after 2022 are a property of the closure, not a forecast; Section 6.6 shows that the demographic results do not depend on the closure.

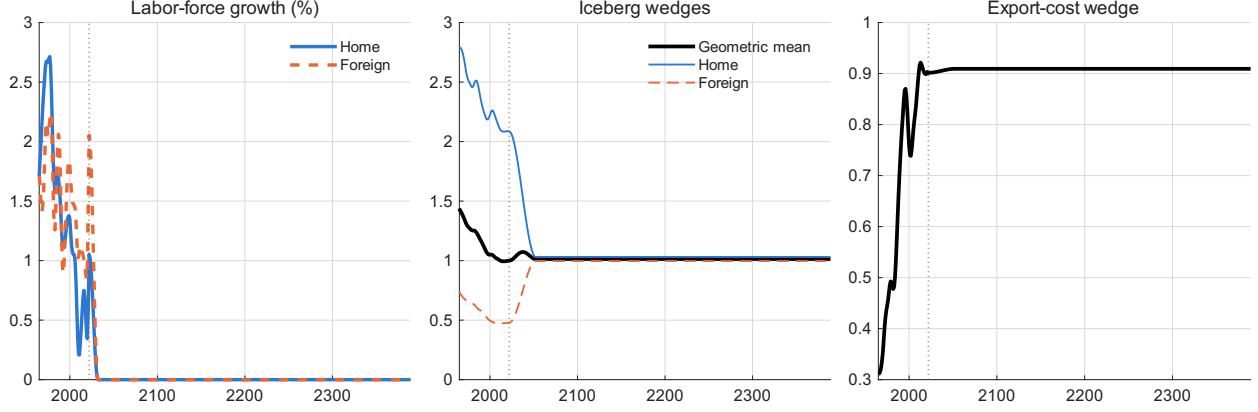


Figure 10: Inputs of the accounting from 1965 to the end of the transition: labor-force growth at home and abroad, the iceberg wedges (geometric mean and the two legs), and the export-cost wedge. The dotted vertical line marks the end of the data in 2022.

E Additional Accounting Results

Two forces. Table 8 groups the iceberg and export-cost wedges into a single force of trade frictions and repeats the Shapley decomposition of Table 3 with two forces. It uses the same solved counterfactuals. Demography’s contribution is nearly unchanged, so it does not depend on how the trade frictions are divided.

Table 8: Shapley decomposition with two forces, 1978–2022.

	Contribution to the path			Contribution to the change (%)		
	Trade frictions	Labor force	All forces	Trade frictions	Labor force	All forces
<i>Panel A. Baseline economy</i>						
Exporter share	0.69	0.31	1.00	30	70	100
Entry rate	−0.04	0.52	0.48	−35	78	43
Exit rate	−0.01	0.23	0.23	−14	43	30
Average size	−0.04	0.21	0.17	−23	25	1
Exports / consumption	0.87	0.13	1.00	83	17	100
Trade balance / consumption	1.18	−0.18	1.00	122	−22	100
<i>Panel B. Symmetric economy</i>						
Exporter share	0.88	0.12	1.00	77	23	100
Entry rate	0.00	0.52	0.52	0	78	78
Exit rate	0.00	0.23	0.23	−1	43	42
Average size	0.08	0.17	0.25	10	20	30
Exports / consumption	0.98	0.02	1.00	98	2	100

Notes: “Trade frictions” are the two iceberg wedges and the export-cost wedge together. Statistics as in Table 3.

The directly observed window. The exporter share before 1992 is constructed from earlier Census benchmarks (Appendix A). Table 9 measures demography’s contributions over 1992–2022, from the same solved paths. Its contribution to the path of participation is 0.32,

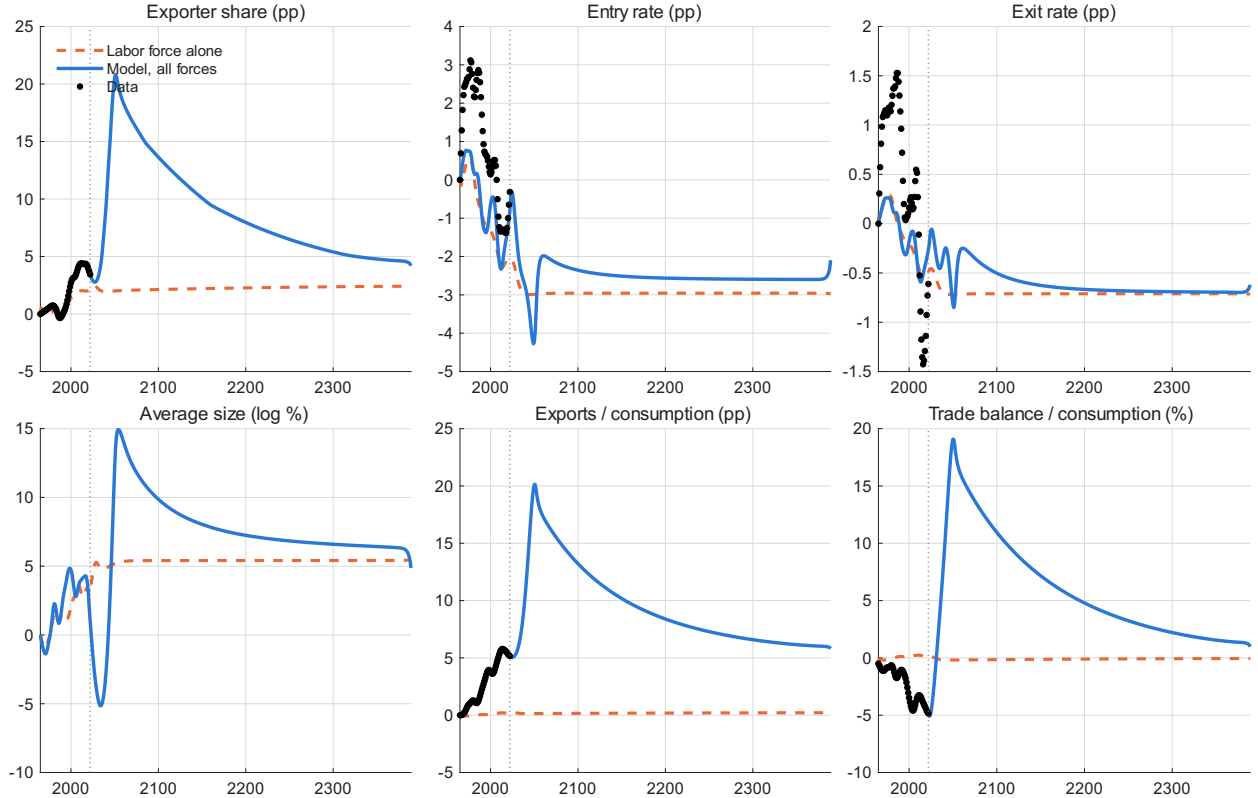


Figure 11: Data (markers), the fitted economy (solid), and labor-force growth alone with every wedge at its 1965 value (dashed), from 1965 to the end of the transition; changes since 1965 except for the trade balance. The dotted vertical line marks the end of the data in 2022.

against 0.32 over 1978–2022, and in the gap economy 0.28, against 0.29. Its contribution to the change in participation is smaller over the shorter window, 43 against 73 percent: demography accounts for a larger part of the rise before 1992 than after it.

Separating the iceberg wedges. Table 10 separates the two iceberg wedges, using the counterfactuals for all sixteen combinations of the four forces. The closure sets the split between the two (Section 6.6), so we report their separate contributions only for completeness. Demography’s contribution is the same as with three forces.

F Calibration Variants

Table 11 compares the 1978 calibration of Section 4 with the one-percent calibration used by the controlled transitions of Section 5. Both target the same 1978 moments and share the assigned parameters except initial labor-force growth and β , which sets the bond price to

Table 9: Demographic contributions over two windows.

	Baseline economy: labor force (Shapley)				Gap economy: labor force only	
	Path		Change (%)		Path	
	1978–2022	1992–2022	1978–2022	1992–2022	1978–2022	1992–2022
Exporter share	0.32	0.32	73	43	0.29	0.28
Entry rate	0.52	0.49	78	91	0.52	0.49
Exports / consumption	0.13	0.17	17	21	0.05	0.05

Notes: Baseline economy: Shapley value of labor-force growth as in Table 3. Gap economy: contribution of labor-force growth alone, wedges held at their 1965 values. “Path” and “Change” as in Table 3, measured over the stated window.

Table 10: Shapley decomposition with four forces, baseline economy, contribution to the path, 1978–2022.

	Home iceberg	Foreign iceberg	Export cost	Labor force	All forces
Exporter share	1.27 [1.18]	0.91 [0.90]	−1.50 [−0.70]	0.32 [0.32]	1.00
Entry rate	0.02 [0.01]	−0.06 [−0.07]	0.00 [0.00]	0.52 [0.52]	0.48
Exit rate	0.01 [0.01]	−0.02 [−0.03]	0.00 [0.00]	0.23 [0.23]	0.23
Average size	0.08 [0.06]	−0.13 [−0.16]	0.01 [0.00]	0.20 [0.18]	0.17
Exports / consumption	0.54 [0.31]	0.46 [0.24]	−0.12 [−0.06]	0.12 [0.05]	1.00
Trade balance / consumption	0.03 [−0.08]	1.16 [1.08]	−0.02 [−0.01]	−0.16 [−0.05]	1.00

Notes: Value of the force acting alone in brackets. Statistic as in Table 3.

0.96 at that growth rate. At one percent growth the model cannot reach the 1978 entry and exit rates, which reflect faster growth (Corollary 1(b)); it matches the export moments as closely as the 1978 calibration does. The one-percent calibration attributes more of exporters’ persistence to the start-up premium ($\kappa_0/\kappa_1 = 1.59$, against 0.99) and less to the persistence of productivity ($\rho_z = 0.937$, against 0.972). The stopper ratio disciplines both, so the targets separate them only weakly (Table 2, notes). Section 5.1 reports the symmetric transition at the 1978 parameters.

G Endogenous Labor Supply

With the preferences of King et al. (1988), the household of country c maximizes

$$\sum_{t=0}^{\infty} \beta^t L_{c,t} \left[\log c_{c,t} - \chi \frac{h_{c,t}^{1+\varphi}}{1+\varphi} \right], \quad \chi > 0, \varphi \geq 0,$$

where $h_{c,t}$ is hours per member of the labor force, subject to (3)–(4) with labor income $W_{c,t}h_{c,t}$ in place of $W_{c,t}$. Utility is additively separable and logarithmic in consumption, so the discount factors $Q_{c,t} = \beta c_{c,t}/c_{c,t+1}$ and the international Euler condition (5) are

Table 11: The 1978 and one-percent calibrations.

<i>A. Parameters</i>			
Parameter	Symbol	1978 calibration	One-percent calibration
Elasticity of substitution	θ	5	5
Discount factor	β	0.966	0.962
Initial labor-force growth (%)	n_0	2.64	1.00
Scale of the operating shock	σ_x	0.001	0.001
Structural iceberg cost	τ	0.78	0.78
Start-up export cost	κ_0	0.0176	0.0304
Continuation export cost	κ_1	0.0177	0.0191
Productivity persistence	ρ_z	0.972	0.937
Productivity innovation s.d.	σ_z	0.126	0.174
Entrant productivity gap	μ_E	0.078	0.015
Operating fixed cost	f_O	0.0270	0.0476
Exogenous exit probability	δ	0.053	0.055
Entry cost (implied)	f_E	1.12	0.90
<i>B. Targeted moments (percent)</i>			
Moment	Data	1978 calibration	One-percent calibration
Exporter share	14.72	14.72	14.73
Stopper ratio	17.00	17.00	17.00
Entry rate	14.20	14.09	13.06
Exit rate	11.31	11.45	12.06
Employment share of entrants	6.80	6.88	6.76
Employment share of exiters	5.27	5.20	5.28
Export intensity of exporters	9.03	9.03	9.03

Notes: Costs and σ_x are in units of labor. Each model column is the balanced growth path at its own n_0 with every wedge equal to one. The size distribution is targeted in both calibrations (Figure 2 shows the fit of the 1978 calibration).

unchanged. Hours satisfy the intratemporal condition

$$\chi h_{c,t}^\varphi c_{c,t} = W_{c,t}, \quad (31)$$

and labor-market clearing (15) becomes $h_{c,t} = \ell_{c,t}^P + \ell_{c,t}^O + \ell_{c,t}^X + f_E m_{c,t}$, with each term in units of labor per member of the labor force.

On a balanced growth path consumption and the wage grow at the common rate G_p (Proposition 1(ii)), so the ratio $W_{c,t}/c_{c,t}$ is constant and, by (31), so are hours. Detrended by X_t , the intratemporal condition reads $\chi h_c^\varphi \hat{c}_c = \hat{W}_c$ and does not involve G_L . Proposition 1 therefore extends with hours appended to the stationary objects: firms discount detrended values at β , as in (17); G_L enters the detrended conditions only through the factor $1/G_L$ in (14); and in a symmetric world the free-entry conditions and Lemma 1 pin the export and

exit cutoffs independently of G_L .

Hours can differ across balanced growth paths. By Corollary 1(c), which holds for shares of hours, the share of hours devoted to entry rises with G_L , which changes production labor, income per member, and, through (31), the hours that households supply. A change in hours changes per-capita entry, which scales the per-capita distribution of firms without changing its shape (Corollary 1(a)). The exporter share, the entry rate, and the exit rate are ratios that depend only on that shape, so on a symmetric balanced growth path they are the same functions of G_L as with inelastic labor supply, and the composition channel is unchanged. Along a transition hours also respond to the wealth effects of the changing labor force; the quantitative results of the paper assume inelastic labor supply.

H Additional Regional Evidence

Table 12 varies the lag of the birth rate used as the instrument in Section 8, for the exporter share and for the entry rate. The estimates change sign across lags, and every first-stage F statistic is below 10, so conventional Wald inference is unreliable (Lee et al., 2022).

Table 12: Exporter share, entry rate, and labor-force growth: instrumental-variables estimates by lag of the birth rate.

Birth-rate lag (years)	Exporter share			Entry rate		
	15	20	25	15	20	25
Labor-force growth	−0.39** (0.16)	−1.07* (0.55)	2.96 (2.17)	0.35*** (0.11)	0.94** (0.46)	−1.49 (1.07)
First-stage F (Kleibergen–Paap)	4.92	3.52	1.82	5.92	3.70	2.37
Observations	243	279	279	252	297	297

Notes: Division-year observations from 1992–2022 for the exporter share and 1991–2023 for the entry rate, division and year fixed effects, standard errors clustered by year. The instrument is the state birth rate the stated number of years earlier, aggregated to divisions with population weights. Sample sizes differ because the birth-rate series does not cover every lag in every year.

Table 13 reports inference that is robust to a weak first stage and to the small number of clusters. Driscoll–Kraay standard errors use a Bartlett kernel with a bandwidth of three years. The wild bootstrap is the wild restricted efficient bootstrap of Davidson and MacKinnon (2010), computed with `boottest` (Roodman et al., 2019) with 9,999 replications, Rademacher weights for the 31 year clusters and the six-point weights of Webb (2023) for the nine division clusters. The Anderson–Rubin test of $\gamma = 0$ is the test of the reduced-form coefficient of the outcome on the instrument; the 95 percent sets invert it over a grid from −10 to 10 in steps of 0.01, and an interval that reaches the end of the grid is reported as

unbounded.

Table 13: Inference for the instrumental-variables estimates.

Birth-rate lag (years)	Exporter share on labor-force growth			Entry rate on labor-force growth	Exporter share on entry rate
	15	20	25	20	20
Estimate	-0.39	-1.07	2.96	0.94	-1.12
First-stage F , year clusters	4.92	3.52	1.82	3.70	32.30
First-stage F , Driscoll–Kraay	3.02	2.52	1.41	2.77	14.19
<i>p-values of the Wald test</i>					
Clustered by year	0.021	0.062	0.182	0.050	<0.001
Driscoll–Kraay	0.045	0.113	0.234	0.071	<0.001
Wild bootstrap, by year	0.021	0.016	0.063	0.004	0.003
Wild bootstrap, by division	0.150	0.153	0.333	0.066	0.112
<i>p-values of the Anderson–Rubin test</i>					
Clustered by year	0.002	<0.001	<0.001	<0.001	<0.001
Driscoll–Kraay	0.041	0.021	<0.001	<0.001	0.021
Wild bootstrap, by division	0.151	0.141	0.135	0.187	0.141
<i>95 percent Anderson–Rubin confidence set</i>					
Clustered by year	$[-4.01, -0.18]$	$(-\infty, -0.45] \cup [9.63, \infty)$	$(-\infty, -5.26] \cup [1.13, \infty)$	$[0.45, \infty)$	$[-1.59, -0.66]$
Clustered by division	$[-1.86, 0.16]$	$(-\infty, 0.55]$	$[-0.39, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$
Observations	243	279	279	297	279

Notes: Division-year observations, division and year fixed effects. The instrument is the state birth rate the stated number of years earlier, aggregated to divisions with population weights. Columns (1)–(3) and (5): 1992–2022; column (4): 1991–2023. The last column instruments the entry rate. Sample sizes differ because the birth-rate series does not cover every lag in every year. Anderson–Rubin sets clustered by division use the conventional critical value with eight degrees of freedom.

Table 14 averages each variable over non-overlapping periods of 1992–2022 (five-year periods, the last of six years; ten-year periods, the last of eleven; and the halves 1992–2006 and 2007–2022) and regresses the period averages on each other with division and period effects. With two periods this is a long difference across divisions. The instrument is the period average of the birth rate twenty years earlier.

Table 14: Period averages: exporter share, entry rate, and labor-force growth.

	Five-year periods		Ten-year periods		Two halves	
	OLS	IV	OLS	IV	OLS	IV
<i>Exporter share</i>						
Labor-force growth	−0.12 (0.11)	−1.01 (0.78)	−0.11 (0.35)	−1.68 (1.43)	−0.12 (0.54)	−1.63 (1.54)
Wild bootstrap p -value	0.111	0.220	0.664	0.241	0.727	0.326
Anderson–Rubin p -value	—	0.110	—	0.048	—	0.157
First-stage F	—	5.13	—	3.36	—	4.05
<i>Entry rate</i>						
Labor-force growth	0.23 (0.09)	0.85 (0.41)	0.22 (0.20)	1.14 (0.69)	0.59 (0.30)	1.17 (0.51)
Wild bootstrap p -value	0.017	0.055	0.274	0.137	0.092	0.057
Anderson–Rubin p -value	—	0.122	—	0.058	—	0.052
First-stage F	—	5.13	—	3.36	—	4.05
Observations	54	54	27	27	18	18

Notes: Division-period averages over 1992–2022, division and period fixed effects. Standard errors in parentheses, clustered by division. Wild-bootstrap p -values (wild bootstrap for OLS, wild restricted efficient bootstrap for IV) and Anderson–Rubin p -values use Webb weights over the nine division clusters, 9,999 replications.